School-Linked Social Insurance and Labor Market Effects: Evidence from Mexico[1]

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July 2026

Abstract

In 2016, Mexico expanded access to the national social insurance system (IMSS) to more than six million public high school and college students, linking medical benefits to school enrollment rather than to formal employment. A central concern with such policies is that delinking benefits from formal contracts may tilt incentives toward informality. I estimate the impact of the high-school component on early labor market outcomes, exploiting variation in exposure across birth cohorts interacted with pre-policy state informality to identify an intention-to-treat differential. Using national labor force surveys, I find that in states with higher pre-policy informality, exposed cohorts exhibit higher labor force participation and formal employment relative to adjacent unexposed cohorts. A one-percentage-point increase in a state's initial informality is associated with a 0.8–1.3 percentage-point wider exposed-minus-unexposed cohort gap in formal employment, concentrated in wage employment, with no accompanying rise in informal employment or unemployment. Effects are similar for women and men, and the timing of the cross-cohort divergence aligns with policy exposure rather than pre-existing trends. The pattern is consistent with reduced informational and administrative frictions to formal hiring (enrollment assigned the permanent social security number required for formal employment) rather than benefit substitution.

[1]The views and conclusions presented in this paper are exclusively of the author and do not necessarily reflect those of Banco de México or its Board of Governors. I am grateful to Nicholas Bloom, Caroline Hoxby, Ran Abramitzky, Aldo Heffner, Alfonso Cebreros, Daniela Puggioni, and two anonymous Banco de México referees for valuable comments. All errors are my own.

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1 Introduction

Informal employment refers to jobs lacking legal and social protection in law or in practice. According to the International Labour Organization, more than 2 billion workers worldwide are in informal employment, representing over 60% of the workforce (ILO, 2018). While informal employment is present across all countries, it is a key characteristic of labor markets in developing countries (Ulyssea, 2020; Breza and Kaur, 2025). High rates of informality are associated with higher incidence of poverty and job insecurity (Ulyssea et al., 2025; Donovan et al., 2023; Brockmeyer et al., 2025). At the macro level, informality represents an important obstacle for tax collection and social insurance provision (Jensen, 2022; Bachas et al., 2023). Informal workers usually lack access to social protection programs that provide a safeguard against negative shocks such as disease or unemployment.

Policymakers and international institutions have raised the question of whether expanding social protections outside formal employment reduces the relative value of formal contracts and thus tilts incentives toward informality (Levy, 2008). This debate has intensified with the global push toward universal social protection (USP2030), led by the World Bank Group and the International Labour Organization. The central question, therefore, is how policy design interacts with pre-existing informality to shape early-career formal attachment and, ultimately, the transition to formality.

In this paper, I study a school-linked social insurance policy enacted in Mexico. In 2016, students enrolled in public high schools and colleges began receiving medical benefits from the national social insurance system (IMSS) while they remained in school, independently of their employment status. Using information from the quarterly National Survey of Occupation and Employment (ENOE), I evaluate the impact of gaining access to these benefits on early career labor market outcomes of the exposed population, varying with initial informality. Early experiences in the labor market are important as there is evidence of high persistence of informality (Akay and Khamis, 2012) and of *scarring* effects of youth informality (Cruces et al., 2012) on future employment and earnings. The Mexican labor market provides a good scenario for this study, as it has a large share of informal employment (close to 60%), with significant heterogeneity across states.[1]

[1]National informality fell modestly from 57% in 2016 to 55% in 2025. Informality at the state level ranges roughly from 40% to 80%. Source: INEGI.

Using a cohort-by-state difference-in-differences design, I compare birth cohorts that were still in school in 2016 (and therefore eligible for school-linked IMSS benefits) with slightly older adjacent cohorts serving as controls, and interact exposure with pre-policy state informality to recover an intention-to-treat differential. Because eligibility was rolled out nationwide within a short window, there is no same-age untreated counterfactual or staggered geographic adoption. The design therefore relies on cross-cohort exposure and pre-existing state conditions rather than a direct treated vs. untreated comparison. Intuitively, the design asks whether the exposure gap between adjacent cohorts is larger where informality was initially higher, net of cohort, state, and time effects.

Exposed cohorts in more informal states display higher labor force participation and formal employment relative to adjacent unexposed cohorts in the same states, compared to that gap in less informal states. Quantitatively, each 1 percentage point increase in initial informality is associated with a 0.4–0.6 percentage-point wider exposed-minus-unexposed gap in participation and a 0.8–1.3 percentage-point wider gap in formal employment. The evidence points to formal sector entry: decomposing labor market status as shares of the cohort population, the entire participation response is absorbed by employment, formal employment absorbs more than the full increase, and informal employment declines, with no movement in unemployment. The formalization response thus combines new entrants taking formal jobs with a reallocation away from informal work, and is concentrated in wage employment, with no response of self-employment. I find no detectable effects on school enrollment, and income effects are statistically detectable in the panel specification but economically small (approximately 1.1 USD per month).

These results are stable and identified. Effects are similar for women and men, for students and non-students, and for individuals in households with formal and informal household heads. Results are robust to excluding the self-employed. Evidence from ENILEMS, a specialized survey of high-school graduates, corroborates the increase in participation and shows qualitatively similar patterns for formalization. Identification is supported by the pattern of cohort-specific informality gradients (tightly estimated, drifting slightly in the direction opposite to the treatment effect among untreated cohorts, and rising sharply only at the exposure ramp) and by placebo versions of the design run on cohorts too old to be treated, which yield null or opposite-signed estimates.

Taken together, the pattern (effects concentrated in formal wage employment, absent in self-employment, and persistent after school exit) is most consistent with the policy's salience and administrative-frictions channel: enrollment made benefits concrete and assigned the permanent social security number required for formal hiring. In this setting, the answer to the delinking concern is therefore negative: expanding health coverage outside formal contracts did not tilt young workers toward informality (informal employment fell where exposure was strongest) and instead strengthened early formal attachment.

The remainder of the paper is organized as follows. In Section 2, I describe the policy and the potential mechanisms. In Section 3, I briefly discuss the relevant literature. Section 4 is devoted to the description of the data, research design, and presentation of the main results. Section 5 presents results by school enrollment, gender, and an alternative definition of informality. Section 6 presents results from an alternative survey, with detailed information on the labor market outcomes of high-school graduates. Section 7 concludes.

2 Expansion of Social Insurance Benefits for Students

In late 2015, Mexico established a health insurance plan for all students enrolled in public high schools and universities. Beginning in 2016, students enrolled in public schools became eligible for IMSS medical benefits while in school, independently of their employment status. Enrollment assigned each student a permanent social security number, which is required to hold a formal job in Mexico.[2] Students obtained free access to medical consultations, lab tests, medicines, certain surgical interventions, pregnancy support, and guidance on nutrition, addictions, and sexual education.

In 2016, the federal and state governments launched the "Tienes IMSS" (You Have IMSS) campaign to enroll students and promote benefit take-up. There was a widespread media campaign and big political events with the presence of the president, cabinet members, and governors of the states announcing the enrollment progress. Between March and August 2016, over six million students were registered, effectively covering the universe of students in public high schools and colleges. Although administrative efforts were coordinated by states and thus staggered, the window from launch to near-universal coverage was about five months. Given the quarterly frequency of ENOE, this rapid nationwide rollout leaves insufficient cross-state timing variation to exploit geographic stag-

[2] The social security number in Mexico is not used as the main identification number, but it is required for formal employment and remains fixed over the life cycle.

gering. The empirical strategy therefore relies on cross-cohort exposure interacted with pre-policy state conditions rather than a direct treated-versus-untreated comparison. Figure A.1 in the Appendix shows how enrollment into the program evolved. Section B in the Appendix includes additional details about the operation of the program.

In the Mexican context, the core distinction between formal and informal employment is access to social insurance (principally healthcare and retirement contributions). Survey evidence indicates that health coverage is the most valued component of formal jobs.[3] Public-school enrollment rates are high: the coverage rate of the population in high-school age is 74% and roughly 81% of high-school students attend public institutions.[4] Thus, this policy affected a substantial portion of Mexican youth and provides a setting to study how early, school-linked access to health insurance (decoupled from holding a formal job) shapes early-career labor market outcomes.

Ex-ante, this policy has an uncertain impact on the early labor market outcomes of the exposed population. I focus on three channels:

1. **Schooling margin (stay in school longer).**
   By attaching medical benefits to current enrollment in school, the policy raises the return to staying in school. If this dominates, exposed students delay labor market entry: lower participation while enrolled, possibly higher enrollment, and ambiguous effects on formality conditional on working.

2. **Benefit substitution while enrolled (relative price of informality).**
   While most high-school students are full-time, a non-negligible minority works alongside school. Using ENOE, I document that among individuals of typical high-school age (15-18) that report going to school, the average labor force participation rate over 2015-2019 is about 14%. Conditional on being part of the labor force, 95% are employed. Consistently, ENILEMS 2019 reports that 26% of recent high-school graduates had some work experience before finishing school. These magnitudes establish that the "study-and-work" margin is empirically relevant. Accordingly, the student IMSS expansion can plausibly reduce the incremental value of formal jobs' benefits while enrolled.

   Formal jobs typically bundle non-wage benefits B^F (IMSS coverage) with wages

[3] *Módulo de Trayectorias Laborales (MOTRAL) 2015.*

[4] Secretaría de Educación Pública. *Principales Cifras del Sistema Educativo Nacional 2015–2016*.

w^F. Informal jobs pay w^I with $B^I \approx 0$. Let $b^S > 0$ denote the value of school-linked benefits while enrolled. For students, the formal premium becomes:

$$(w^F - w^I) + (B^F - b^S).$$

If b^S appreciably closes the benefit gap $(B^F - b^S)$, the relative attractiveness of informal work rises. Among the currently enrolled, we could expect a reallocation toward informal employment. Effects should dissipate after leaving school (when $b^S = 0$). Wage effects are a priori ambiguous, as employers may substitute wages for benefits. This is precisely the concern raised by policymakers and international institutions when expanding social protection outside formal employment: delinking benefits from contracts may unintentionally subsidize informality at the margin.

3. **Salience, rights, and reduced frictions toward formality during the school-to-work transition.**
The program made IMSS benefits concrete (healthcare) and assigned a permanent social security number, lowering informational and administrative frictions to formal hiring. If students learn and value these benefits, they may seek formal jobs to retain them after school, and employers face lower onboarding frictions. We could expect higher formal employment at exit (through both entry into the labor force and reallocation away from informal jobs) concentrated in wage employment rather than self-employment, with no rise in informality, and stronger effects where pre-policy informality was high.

Given these opposing forces, the net effect is empirical.

3 Literature Review

This paper is most closely related to the literature studying the relationship between social protection programs that are not attached to employment and labor market outcomes, where there is a debate of their impact. In developed countries, the effect of welfare programs on labor supply has been extensively studied, with most evidence drawing on means-tested transfers and income support schemes finding negative effects on work incentives (Moffitt, 2002; Meghir and Phillips, 2008). More related to this paper, Akosa Antwi et al. (2013) find that the Affordable Care Act's expansion of dependent coverage until age 26 resulted in fewer hours worked and a lower likelihood of working full-time jobs. For Latin America, the evidence on conditional cash transfer (CCT) programs points to negative but small and insignificant effects on the labor supply of the targeted population (Alzúa et al., 2013). However, they point to differentiated effects across genders and within the household. Alternatively, Gerard et al. (2021) find that an expansion of *Bolsa Familia* in Brazil increased local formal employment.

In the case of Mexico, there have been several studies that focused on the impact of the *Seguro Popular* program, which provided medical benefits to poor households. On one side, one of the main exponents of the argument that certain forms of social insurance can cause substantial increases in informality is Levy (2007; 2008). He argues that *Seguro Popular* acted as an implicit subsidy to informal employment and resulted in a significant reduction of formal jobs. Consistent with this, Bosch and Campos-Vazquez (2014) find evidence of a reduction in employers and employees formally registered. On the other hand, other authors argue against this idea, and most of the empirical literature does not find a significant impact on informality (Esquivel and Ordaz-Díaz, 2009; Arias et al., 2010; Campos-Vázquez and Knox, 2013; Azuara and Marinescu, 2013; Alonso-Ortiz and Leal, 2018). More recently, Seira et al. (2023) use administrative data and improved econometric methods, and find no evidence of a decrease in formal employment. This paper contributes to the literature by analyzing a social program that is not linked to employment but that could affect the incentives toward informality at a critical period, the start of labor market trajectories.

Additionally, this paper is related to a strand of literature that emphasizes the importance of early labor market experiences. There is widespread evidence regarding *scarring* effects of youth unemployment spells, otherwise known as "unlucky cohorts" that face adverse conditions at labor market entry (Arulampalam, 2001; Bell and Blanchflower, 2011; Schwandt and von Wachter, 2019). In the context of Latin America, Cruces et al. (2012) find that youth informal job spells have an impact on persistent informality as well as wage penalty effects. Furthermore, Berniell et al. (2023) provide evidence of gender differences in the region, with women from "unlucky cohorts" showing a higher employment rate and earnings in the future. This result is consistent with the theory that women act as secondary workers in hard times, also known as the "added worker effect" in the literature (see Guner et al., 2025 for a thorough review of this literature). Particularly important for my study is the consideration of working while in school. Le Barbanchon et al. (2023) provide some experimental results of employment lotteries in Uruguay. They find a positive income effect two years after the program, while previous non-experimental evidence provided mixed results (Ruhm, 1997; Hotz et al., 2002; Ashworth et al., 2021).

It is also relevant to note that there are two different traditional schools of thought with respect to the barriers to entry to formal employment. The traditional, dualist school of thought argues in favor of segmented labor markets, based on the Harris-Todaro model (1970) and considers informal workers as being informal in an involuntary way. Alternatively, a revisionist school of thought argues in favor of integrated labor markets, where informality arises from two distinct forces: formal firms choosing to hire workers off the books to reduce labor costs (Portes et al., 1989), and entrepreneurs who optimally choose to operate informally (Maloney, 2004; Bosch and Maloney, 2010). The voluntary interpretation applies most naturally to the latter group; workers hired off the books by formal firms may still face informal status involuntarily. Ulyssea's work (2018; 2020; 2025) has considerably expanded our conceptual and empirical understanding of informality. He distinguishes two margins of informality: the *extensive margin*, where firms decide whether to register formally or not, and the *intensive margin*, where a formal firm can hire informal workers by keeping them "off the books".

For my conceptual framework, it is important to consider the *intensive margin* of informality. My setting highlights the worker's choice between formal and informal jobs at labor market entry. School-linked benefits change two features during enrollment: a benefit substitution component (some medical coverage is available regardless of formal status), which could lower the relative cost of informal work while in school; and a salience/administrative-frictions component (benefits become concrete and a permanent social security number is assigned), which can ease formal hiring at exit. Consistent with the relevance of the intensive margin in Mexico, formal firms post both formal and informal vacancies (Samaniego de la Parra and Fernández Bujanda, 2024).

Accordingly, I test three predictions: (1) effects confined to currently enrolled youth, dissipating at school exit, would indicate benefit substitution; (2) formal-sector entry concentrated at the school-to-work transition (employment expanding through formal wage jobs, with no rise in informal employment) is consistent with salience and reduced frictions; and (3) results should be stronger where pre-policy informality was higher. The evidence below aligns with (2) and (3): total employment among exposed cohorts expands entirely through formal wage employment while informal employment declines, the opposite of what net substitution toward informality would produce, and the effects persist among recent graduates rather than dissipating at school exit. One heterogeneity pattern nonetheless suggests that the substitution margin may operate for students who combine work and school, without generating a net shift toward informality.

4 Impact on Labor Market Outcomes

4.1 Data

The main data source for this paper is the Mexican quarterly survey on employment called ENOE (*Encuesta Nacional de Ocupación y Empleo*).[5] This survey has information on approximately 200 thousand workers each quarter, is representative at the national and state level, and allows me to have information for both formal and informal workers. The design of the survey is a rotating panel, such that each individual is followed for 5 quarters, and each quarter, one fifth of the sample is replaced. Crucially, ENOE allows me to implement INEGI's official definition of employment informality, ensuring a consistent and comparable measurement across time and states. I adopt INEGI's employment-informality concept (worker/job status) and construct outcomes using the pre-coded survey variables.[6]

The cleanest design would randomize (or stagger exogenously) access to school-linked IMSS benefits across schools or states and track named individuals longitudinally through school exit and early careers, observing school type (public/private), eligibility, take-up and subsequent entry into the labor market. In practice, the 2016 rollout reached the universe of eligible public-school students nationwide within about five months, leaving no exploitable same-age untreated group or multi-period geographic staggering at ENOE's quarterly frequency. Moreover, ENOE's short rotating panel does not support tracking individuals from school to work at scale, and it does not identify whether a respondent attended a public vs. private high school nor student-IMSS take-up. I therefore adopt a cohort-by-state difference-in-differences approach: compare birth cohorts exposed to the 2016 program (still in school at rollout) with slightly older adjacent cohorts, and interact exposure with pre-policy state informality to recover a differential intention-to-treat effect. Intuitively, I ask whether the exposure gap between adjacent cohorts is larger in states that were initially more informal. The main analysis centers on 2019, when the first

[5] Due to the COVID-19 pandemic there is no information for the second quarter of 2020. For the period between the third quarter of 2020 and the fourth quarter of 2022 the information comes from the ENOE New Edition ($ENOE^{N}$). This temporary survey maintains the conceptual, statistical, and methodological design of ENOE, but combines in-person interviews with phone interviews. Throughout this period, in-person interviews represent between 79% and 100% of all interviews in $ENOE^{N}$.

[6] Specifically INEGI's definition of informal employment considers the following groups: self-employed and independent workers in the informal sector (economic units operating with household's resources that are not identifiable and independent of those households), self-employed workers in the agricultural sector, workers without pay, domestic workers without social insurance, employees with wages or non-wage perceptions but no social insurance, and workers with non-specified occupation position.

set of fully exposed cohorts completed high school and before the COVID-19 shock.

I complement my main results with information from a survey that measures the labor market insertion of high-school graduates (aged 18–20), called ENILEMS (*Encuesta Nacional de Inserción Laboral de los Egresados de la Educación Media Superior*). This survey provides more detailed information about an individual's high school and labor market experiences. Crucially, I can identify whether they attended a public or private high school, making the identification of exposure to the policy cleaner. ENILEMS also provides information on the educational attainment of parents, whether the individuals attempted to, and were ever enrolled in a higher education institution, and whether the individuals were employed during high school. Unfortunately, this survey is not administered regularly. However, I can use the 2016 and 2019 surveys (the two most recent ones), that capture outcomes for cohorts both exposed to and unaffected by the policy. It is important to mention that ENILEMS started as a special module of ENOE and for its 2019 wave it became an independent survey but the sampling follows households identified by ENOE, making both surveys fully comparable. Given the limited number of waves and smaller sample, ENILEMS plays a corroborating role: it sharpens the exposure proxy and enables mechanism checks, but it is not the basis for identification.

4.2 Research Design

I focus on the high-school component of the policy (roughly U.S. grades 10–12) for three reasons. First, the mapping from birth cohorts to school grade/age is tight: most high-school students are 15–18, so cohort of birth is an informative proxy for exposure. Second, coverage at high-school ages is high (74% of the age group enrolled, 81% in public schools), whereas college enrollment in Mexico is approximately 30% with much greater age dispersion.[7] Focusing on high school therefore reduces composition noise and improves interpretability. Third, the 2016 rollout reached universal coverage of public high-school students within five months, leaving too little geographic timing variation at ENOE's quarterly frequency to credibly exploit staggered adoption. For these reasons, the birth cohort determines eligibility exposure in my design. Analyzing the college component and longer-run outcomes is left for future work.[8]

[7] Secretaría de Educación Pública. *Principales Cifras del Sistema Educativo Nacional 2015-2016*.

[8] Because the policy also applied to public universities, some individuals in the control cohorts (those just older in 2016) could have become eligible upon entering public college. ENOE does not identify public versus private school attendance, so I cannot fully purge this "treated-controls" contamination, which is quantitatively relevant: 39% of the 1995–1998 cohorts are observed attending school during 2016–2019 (Appendix Table A.7). This partial treatment of the comparison group attenuates cohort contrasts, but the

The ideal experiment would randomly assign school-linked IMSS eligibility at the individual (or school) level and follow named individuals before and after the reform, comparing treated and otherwise identical controls of the same age, state, and background. With such data, we could estimate a direct treated vs. untreated effect within an age cell and trace dynamics through school exit and early careers.

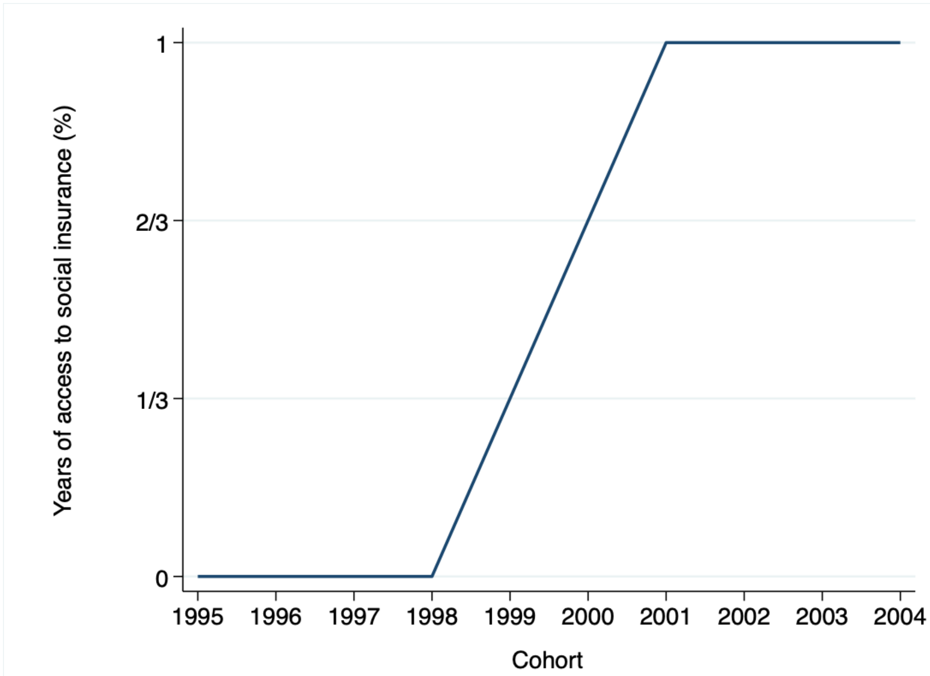
However, eligibility was expanded nationwide and almost simultaneously in 2016. Any staggered administrative adoption across states was compressed into 5 months, too short to exploit at ENOE's quarterly frequency. ENOE also does not record public vs. private high school (nor student-IMSS take-up), and its rotating panel does not follow youth long enough to observe clean transitions from school to stable employment.[9] Crucially, for the young treated cohorts we do not observe pre-policy labor outcomes because they were below labor force age before 2016. These features rule out a same-age treated vs. control comparison and a conventional before-after design.

I therefore exploit the eligibility cutoff in cohort space. I measure exposure continuously as the share of grades 10–12 that coincided with the implementation of the policy. Figure 1 plots the exposure measure. Cohorts aged 18 or older in 2016 (born in 1998 or earlier) had already graduated from high school before the policy and are fully unexposed. Cohorts aged 15 or younger in 2016 (born in 2001 or later) entered high school after full implementation and are fully exposed. The two intermediate cohorts (born in 1999 and 2000, aged 17 and 16 in 2016) are partially exposed, having had access to benefits for one or two years of high school, respectively. The long-difference specification in Section 4.3.1 compares only the fully exposed cohorts with the fully unexposed cohorts. The panel specification in Section 4.3.2 additionally incorporates the partially exposed cohorts through the continuous exposure measure.

attenuation is not uniform: contamination declines with initial informality at an estimated rate of 0.205 percentage points per point of informality (Table A.7 Panel B), which by itself would inflate the estimated gradient. Under the same premise that motivates the design (that schooling-linked effects scale with initial informality) the net contribution of contamination to the estimated slope is proportional to $-[c_s - 0.205 \cdot I_s]$, so attenuation dominates whenever a state's contamination level exceeds 0.205 times its informality rate. This holds in every state by a wide margin: the minimum contamination (27.7%) exceeds the maximum threshold ($0.205 \times 80 = 16.4\%$). The estimates may therefore be read as lower bounds on the combined high-school-plus-college schooling-linked impact, conditional on college eligibility affecting outcomes in the same direction as high-school eligibility; if the college effect were instead constant across states, contamination would impart a small upward bias bounded by 0.205 times that effect. Section 5.4 quantifies the implied magnitudes per enrolled-eligible student.

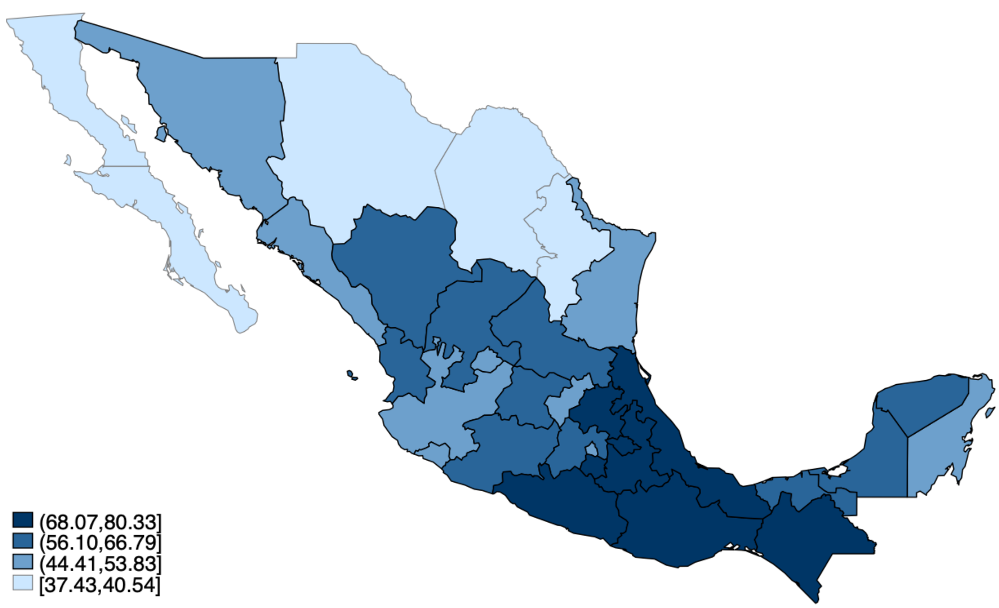
[9] Even if school type were observed, public-private comparisons would not necessarily be credible counterfactuals: arguably they could differ systematically in selection, resources, and trajectories.

Figure 1: Exposure to high-school-linked social insurance by cohort



Note: The figure shows the exposure to the policy by cohort, defined as the percentage of years of high school when students had access to social insurance benefits.
Source: Author's calculation.

Figure 2: Pre-policy informality rate by state (%)
2013–2015 average



Source: Author's calculations with data from ENOE.

Because the reform's bite plausibly depends on local frictions and returns to formality, I interact this cohort exposure with pre-policy state informality. Figure 2 plots the pre-policy informality rate by state (2013–2015 average), showing substantial heterogeneity. I treat the pre-policy informality as an intensity shifter of the reduced-form effect, which I am unable to estimate in the conventional before-after design. Intuitively, the design asks whether the gap between adjacent cohorts is larger in states that were initially more informal, after absorbing cohort, state, and time effects, as well as common shocks.

This strategy identifies an intention-to-treat (ITT) differential. That is, how the policy's effect scales with initial informality, rather than an aggregate average effect or a within-age treated vs. untreated effect. This mirrors the logic in Duflo (2001) and Bleakley (2010): exploit timing by cohort and variation in pre-policy intensity, and demonstrate that only the exposed cohorts exhibit breaks aligned with exposure.

Outcomes are constructed at the cohort-state-quarter level. Intuitively, the difference-in-differences (DiD) design is outlined in Table 1. For simplicity, consider two state groups by pre-policy informality: low (L) and high (H). Within each group compare exposed cohorts (still in high school at rollout) to unexposed adjacent cohorts. Let Y_g^E and Y_g^U denote mean outcomes for group $g \in \{L, H\}$. Each $\Delta_g = Y_g^E - Y_g^U$ is the exposed-unexposed gap within a given state environment. The difference-in-differences estimand across state groups is $\beta = \Delta_H - \Delta_L$. This is the ITT differential: how the exposure effect varies when moving from low to high pre-policy informality. In the continuous specification used in the paper "L vs. H" becomes the slope of Δ_s with respect to $I_{s,pre}$ (pre-policy informality at the state level), and exposure is measured continuously (share of grades 10-12 under the policy).

Table 1: Cohort-by-state DiD with intensity

	Low pre-policy informality	High pre-policy informality
Exposed cohorts	Y_L^E	Y_H^E
Unexposed adjacent cohorts	Y_L^U	Y_H^U
Within-state cohort difference	$\Delta_L = Y_L^E - Y_L^U$	$\Delta_H = Y_H^E - Y_H^U$
Estimand (ITT differential): $\beta = \Delta_H - \Delta_L$		

The unexposed controls are older than exposed cohorts, so any comparison $Y^E - Y^U$ mixes (i) a life-cycle/cohort difference and (ii) the policy effect on the younger, exposed cohort. The design removes the life-cycle component by differencing across state environments.

Let D_s denote the baseline cohort/age gap (unrelated to the policy) and τ_s denote the policy effect on exposed cohorts within state s. We can express the within-state cohort difference as:

$$\Delta_s = D_s + \tau_s.$$

We only observe the sum. However, taking the difference between high- and low-pre-policy informality environments:

$$\beta = \Delta_H - \Delta_L = (D_H - D_L) + (\tau_H - \tau_L)\,.$$

If, absent the policy, the baseline cohort/age gap D_s is the same across state environments (or, more generally, uncorrelated with pre-policy informality after conditioning on fixed effects and controls), then $D_H - D_L = 0$ and the DiD estimand identifies $\tau_H - \tau_L$: the differential policy effect. Equivalently, we allow exposed cohorts to be younger than unexposed, we only require that age/cohort profile of outcomes does not systematically differ by pre-policy informality once we net out cohort, state, and time effects and standard controls. That is the parallel trends in cohort space assumption necessary for identification of the ITT differential. In other words, the identifying assumption is that, in the absence of the policy, the exposed-minus-unexposed cohort gap D_s would have been the same across states with different pre-policy informality—that is, $D_H = D_L$. Only then does $\beta = \Delta_H - \Delta_L$ recover $\tau_H - \tau_L$, the differential policy effect. After conditioning on cohort, state, and time fixed effects and controls, the cohort/age profile of outcomes must not co-vary systematically with pre-policy informality in the absence of the reform.

There are several features of my empirical design that make this assumption plausible. First, I use slightly older adjacent cohorts as controls, minimizing life-cycle spacing. Second, cohort fixed effects absorb national age/cohort profiles; state fixed effects absorb time-invariant differences; time fixed effects absorb national shocks. I consider alternative specifications that include state-level trends to consider differentiated shocks. Third, I estimate β_k (slope of outcome vs. pre-policy informality for each birth cohort) and overlay exposure timing. Identification requires β_k flat for unexposed cohorts and breaking only when exposure is positive. Fourth, I consider placebo cohorts. For cohorts too old to be exposed, $\Delta_H - \Delta_L$ should be zero. Results are robust across gender, school attendance, formality of the household head, and focusing on workers excluding the self-employed.

4.3 Main Results

4.3.1 Long Differences

First, I simplify the analysis and aggregate the different cohorts into two groups: those who were fully exposed to the policy and those who were not exposed at all. I compare the main labor market outcomes (participation rate, employment rate, formality rate, monthly income) across these groups of cohorts for each state to obtain long difference estimates. Therefore, the basic equation to be estimated is:

$$Y_{s,E} - Y_{s,U} = \beta I_{s,pre} + X_{s,pre}\Gamma + \alpha + \varepsilon_s,$$

where Y is a labor market outcome of state s, and the cohort subscripts E/U denote fully exposed and fully unexposed cohort groups, not time periods. In the estimation, outcomes are computed as cohort-state-quarter cell means; the equation is estimated on 32 states $\times$ 4 quarters = 128 observations, with quarter fixed effects δ_t included to absorb seasonal variation. The preceding informality rate is $I_{s,pre}$, defined as the average state informality rate between 2013 and 2015. The X variables are a set of controls measuring conditions at the state level before the implementation of the policy, and α is a constant term. The main outcomes of interest are the participation, employment, and formality rates, and monthly labor income. Throughout, the participation rate is the share of the cohort population in the labor force; the employment rate is the share of the labor force participants who are employed; and the formality rate is the share of the employed holding formal jobs. Because the latter two are conditional rates, movements in them do not by themselves reveal how a participation response is absorbed.[10]

For the set of controls I include state GDP per capita in 2013, mean educational attainment in 2015 and regional controls, grouping the 32 states into 8 different regions, outlined in figure A.2 of the Appendix (see Table A.1 for descriptive statistics). The equation is estimated using weighted least squares with data from the four quarters of the employment survey of 2019. Weighting proceeds in two layers. Cell-level rates and incomes are computed using ENOE expansion factors, so each cell estimates its population quantity. Regressions then weight cells by the square root of cell population, the efficient weighting when cell means are estimated with sampling variance inversely proportional to cell

[10] Table A.2 in the Appendix complements the main results with a full decomposition of labor market status as shares of the cohort population (formal employment, informal employment, unemployment, and non-participation, which sum to one by construction) and splits formal and informal employment into wage employment and self-employment.

size. Results are virtually unchanged using population weights or no weights (Appendix Table A.3). Following Solon et al. (2015), this stability also indicates that the estimates are not driven by heterogeneity across differently sized states. The coefficient β is the ITT differential in long differences: the extra exposed–unexposed gap associated with a 1-p.p. increase in initial informality.

Table 2: Cross Cohort Long Differences in 2019

	(1) Participation rate	(2) Employment rate	(3) Formality rate	(4) Monthly income
Panel A				
Pre-policy informality rate	0.50***	0.16***	0.79***	29.87***
	(0.05)	(0.02)	(0.04)	(3.88)
	[<0.001]	[<0.001]	[<0.001]	[<0.001]
Panel B				
Pre-policy informality rate	0.59***	0.15*	0.87***	11.18
	(0.09)	(0.06)	(0.16)	(10.00)
	[<0.001]	[0.053]	[0.003]	[0.278]
Log GDP per capita in 2013	6.04***	1.58	1.91	-134.53
	(0.92)	(0.99)	(1.33)	(128.86)
Educational attainment in 2015	-3.29***	-0.37	2.25	-328.57**
	(1.10)	(0.82)	(1.88)	(137.75)
Regional controls	✓	✓	✓	✓
Quarter FE	✓	✓	✓	✓
Mean dependent variable	-37.89	0.08	-32.75	-1,930.18
Observations	128	128	128	128

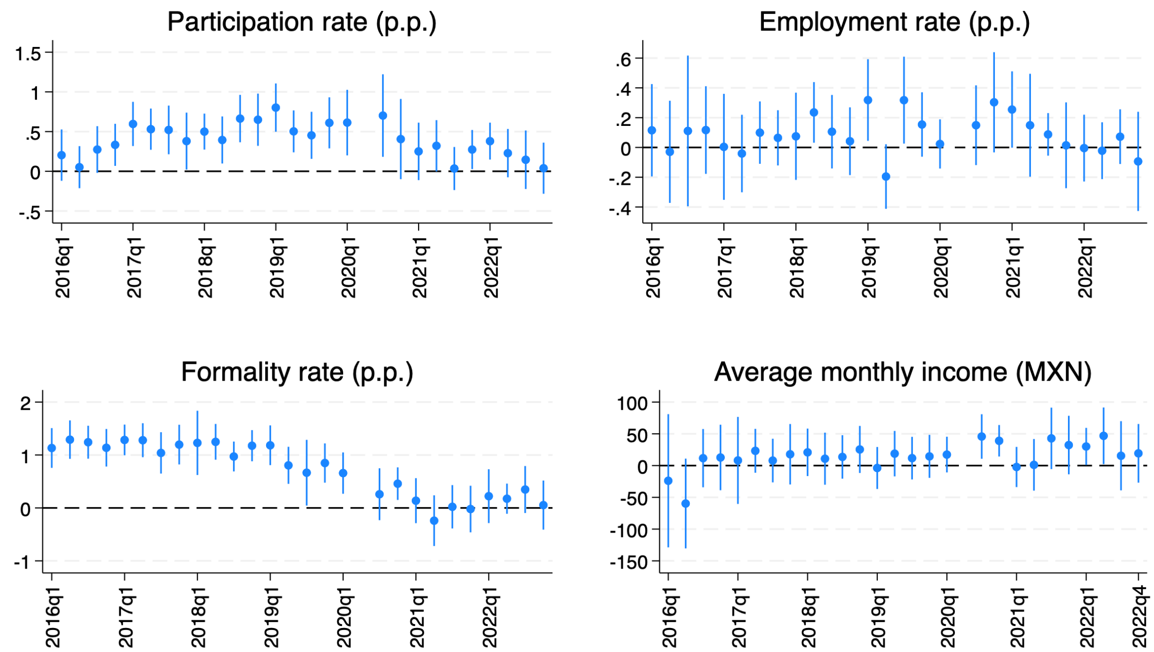
Note: Each column reports a weighted least squares regression of the 2019 exposed-minus-unexposed cohort gap on the state's pre-policy informality rate on 32 states $times$ 4 quarters $=$ 128 state-quarter cells. Cells are weighted by the square root of the cell population (the square root of the positive-income cell population in column (4), which restricts to individuals reporting strictly positive income). Standard errors clustered at the state level in parentheses; wild cluster bootstrap p-values for the pre-policy informality coefficient in brackets (Webb weights, $9,999$ replications; Cameron et al., 2008; Roodman et al., 2019). Stars on the pre-policy informality coefficient are based on the wild-bootstrap p-values. Appendix Table A.3 shows that inference is unchanged under a block bootstrap resampling states, cluster-jackknife (CV3) standard errors, and a specification collapsed to one observation per state.
***p<0.01, **p<0.05, *p<0.1

The long differencing estimates are presented in Table 2. Panel A shows the estimates with the preceding informality rate as the only independent variable, and Panel B includes the full set of controls. From the mean differences between cohorts fully exposed and not exposed at all, we can see that the exposed cohorts have significantly lower participation, and formality rates, as well as average income. These patterns are expected as fully exposed cohorts are younger. Because the treatment-intensity variable varies only across states, the effective number of independent units is 32 regardless of the number of individuals. I therefore cluster standard errors at the state level and, given the modest number of clusters, base inference for the coefficient of interest on wild-cluster bootstrap p-values with Webb weights and 9,999 replications (Cameron et al., 2008; Roodman et al., 2019). Appendix Table A.3 shows that the main estimates are insensitive to the inference method: cluster-robust, block-bootstrap (resampling states), cluster-jackknife (CV3; MacKinnon et al., 2023), and wild-bootstrap intervals all exclude zero by wide margins for participation and formality in both research designs, as does a specification collapsed to one observation per state.

From the estimated coefficients in panel B, one additional percentage point in the preceding informality rate is associated with a wider exposed-minus-unexposed cohort gap of 0.59 percentage points in the participation rate, 0.15 percentage points in the employment rate, and 0.87 percentage points in the formality rate. The estimated impact on labor income, while positive, is not statistically significant. The sample of income results is smaller due to higher non-response rates in this question of the survey. These reduced-form results suggest that in states with higher informality, the exposed cohorts pulled further ahead of the unexposed cohorts in participation and formality, relative to the analogous gap in less informal states.

The analysis is done with information from 2019 to evaluate the medium-term effects of the policy for the fully exposed cohorts in 2016, right before the onset of the COVID-19 pandemic. To consider the full picture, Figure 3 plots the estimated coefficients ($\hat{\beta}_t$) of the long differences equation for each quarter t from 2016 to 2022, keeping the same exposed and unexposed cohort groups throughout. Two caveats apply: as cohorts age over this window the comparison captures increasingly medium- and longer-run effects rather than a fixed early-career snapshot, and some control cohorts may have gained partial exposure through public college after 2016. Pandemic-related disruptions after 2019 introduce an additional break in comparability. The 2016–2019 portion of the figure is therefore the most directly comparable to the long-difference and panel estimates below. Within that window, we observe that the positive effects on the participation and formality rates are relatively consistent. After the COVID-19 pandemic, the estimated coefficients for both of these indicators are close to zero and no longer statistically significant.

Figure 3: Estimated coefficients of pre-policy informality rate on cross cohort long differences, $\hat{\beta}_t$



Note: Each point plots the estimated coefficient from quarterly cross-cohort long difference regressions (exposed minus unexposed) on pre-policy state informality (2013–2015). No available data for 2020q2.
Source: Author's calculations with data from ENOE.

It is not clear how to interpret these estimates as the pandemic severely disrupted labor markets. There is evidence that in Latin America there was an initial drop in participation, employment, and informality (followed by a rebound in the latter), with the informal sector acting as an important margin of adjustment (Alvarez and Pizzinelli, 2021; Leyva and Urrutia, 2023). Furthermore, Osuna-Gómez (2023) finds that people that entered the labor market after the Great Recession lost formal employment during the pandemic at higher rates than previous cohorts. It could be the case that the pandemic shock reverted any effects that were driven by the school-linked social insurance policy on labor market outcomes. However, there are additional concerns for more recent estimates as we are tracking cohorts at the state level and not individuals, and we could worry about potential migration patterns and readjustments between different local labor markets. The employment and income estimates are mostly small and not significant.

4.3.2 Panel Estimates

The long difference estimates transparently summarize medium-run gaps between fully exposed and unexposed adjacent cohorts across states with different initial informality. Their limitation is that they do not, by construction, demonstrate that such gaps were not already drifting with pre-policy conditions. To address this, I turn to a panel framework that (a) uses all quarters from 2016 to 2019 to absorb common shocks with survey-wave fixed effects and (b) follows Bleakley's cohort-space logic: for each birth cohort I recover the cross-state gradient with respect to pre-policy informality and show that this gradient is within a narrow band for cohorts too old to be exposed and turns on only at the eligibility cutoff. In addition, I report robustness with state-specific linear time trends to soak up differential growth by state unrelated to the policy.

First, to incorporate cohorts that were partially exposed to the policy I estimate pooled regressions of the form:

$$Y_{kst} = \tilde{\beta} I_s \times \mathrm{Exp}_k + \delta_k + \delta_s + \delta_t + \sum_i \left(x_s^i \times \mathrm{Exp}_k \right) \gamma_i + \nu_{skt},$$

where Y_{kst} is the labor market outcome of interest (participation rate, employment rate, formality rate, monthly income) for cohort k in state s and quarter t, Exp_k is exposure to the policy for cohort k as defined by figure 1, I_s is the pre-policy informality rate in state s, the x_s^i are state-specific controls, and $\delta_k, \delta_s, \delta_t$ are fixed effects for cohort, state, and quarter, respectively.

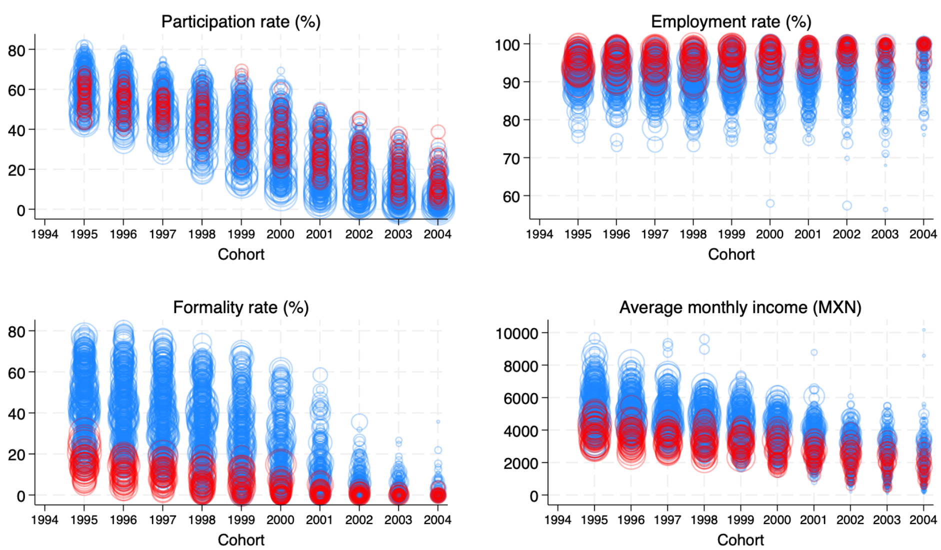
Second, I compare the evolution of labor market outcomes of interest by cohort across states to assess the contribution of the policy to the observed changes. The regressions to be estimated are:

$$Y_{kst} = \beta_k I_{s,pre} + \delta_k + X_s \Gamma_k + \nu_{skt},$$

where β_k is a cohort specific coefficient on pre-policy informality, X_s is a vector of state controls, and δ_k and Γ_k are cohort-specific intercept and slope coefficients. By estimating the equation through weighted least squares for each cohort, I generate a series of estimates across cohorts. The series of cohort estimates helps to show graphical evidence that the results are capturing the impact of the policy by comparing it with the pattern of the exposure measure.

Before presenting the estimates of the pooled and cohort-specific regressions, figure 4 plots the summary statistics for the panel constructed by cohort in states and quarters between 2016 and 2019. Each observation represents the value of the dependent variable for each cohort-state-quarter tuple. The size of the observations is determined by the size of the population represented. I single out in red the three poorest states (Guerrero, Oaxaca, and Chiapas), which also happen to be the southernmost states.

Figure 4: Summary statistics of panel by cohort across states and quarters, 2016–2019



Note: Each observation represents the value of the dependent variable for each cohort by state-quarter. The red observations denote the three poorest states (Guerrero, Oaxaca, and Chiapas). The size of the observations is determined by the size of the population for each cohort-state-quarter tuple.
Source: Author's calculations with data from ENOE.

We observe expected patterns in the data, such as lower participation rates and lower formality rates for younger cohorts. Among the poorest states it is interesting to observe that for the oldest cohorts analyzed, the participation rate is lower relative to richer states but for the youngest cohorts, relative to richer states, the participation rate is higher. It is also worth noting that the employment rate is consistently higher for the poorest states across all cohorts, but the quality of jobs within these states is worse, as seen by the lower formality rate and lower monthly income.

Table 3 presents the estimates for the pooled regressions. The regressions are estimated through weighted least squares with the weights constructed from the square root of cell size. Standard errors are clustered at the state level. These estimates are consistent with the results of the long differencing. The magnitude of the estimate for the participation rate decreased but remains significant. The estimate for the employment rate is very similar, whereas the estimate for the formality rate increased to almost 1.3 percentage points associated with an increase of 1 percentage point in the initial informality rate. This means that taking into account the panel structure and the partial exposed cohorts increases the differential positive impact of the policy on the formalization of exposed cohorts in high informal states relative to low informal states. Income effects are statistically detectable but economically small ($\approx$ 1.1 USD per month), consistent with a reallocation toward formal jobs without large short-run gains in earnings.

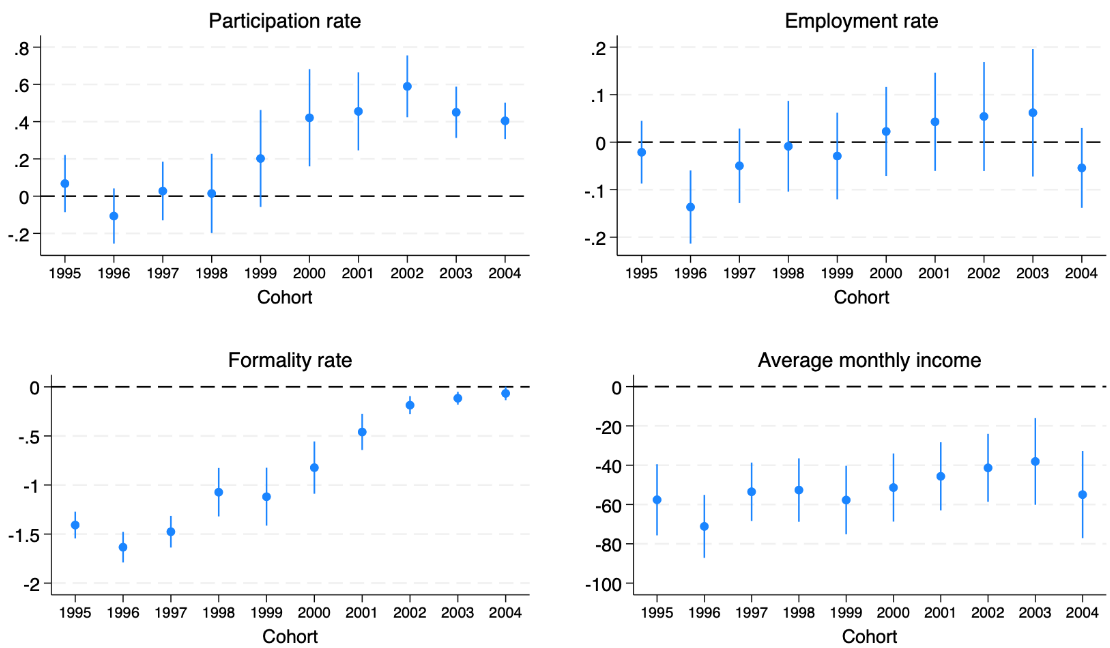
Table 3: Panel Estimates of the Effect of Exposure to Policy, 2016–2019

	(1) Participation rate	(2) Employment rate	(3) Formality rate	(4) Monthly income
Pre-policy informality	0.44***	0.15***	1.28***	21.33**
rate $\times$ Exposure	(0.05)	(0.03)	(0.06)	(7.05)
	[<0.001]	[0.001]	[<0.001]	[0.021]
Log GDP per	2.01	0.43	3.67**	-61.73
capita in 2013 $\times$ Exposure	(1.01)	(1.02)	(1.01)	(114.12)
Educational attainment	-0.33	1.86**	1.86	-72.11
in 2015 $\times$ Exposure	(0.67)	(0.55)	(1.22)	(86.32)
Cohort FE	✓	✓	✓	✓
State FE	✓	✓	✓	✓
Quarter FE	✓	✓	✓	✓
Mean dependent variable	33.31	93.84	17.99	4,299.21
Observations	5,094	5,091	5,091	5,046

Note: Standard errors clustered by state in parentheses. Cells are weighted by the square root of the cell population (the square root of the positive-income cell population in column (4), which restricts to individuals reporting strictly positive income). Wild cluster bootstrap p-values for the informality-by-exposure coefficient in brackets (Webb weights, 9,999 replications).
***p<0.01, **p<0.05, *p<0.1

A natural concern is that labor-demand fluctuations (national or state-specific) could differentially affect youth and mimic treatment heterogeneity. The panel specification mitigates this along several layers, and I add targeted robustness checks. Time fixed effects absorb nationwide demand movements (e.g. macro cycles, fuel-price spike), ensuring identification does not come from aggregate time shocks. State fixed effects soak up permanent gaps in demand, institutions, and composition across states (industrial structure, enforcement). Cohort fixed effects net out nationwide age/cohort profiles in youth demand or school-to-work transitions that are common across states. The identifying variation compares exposed vs. unexposed adjacent cohorts within the same state and quarter, scaled by the state's pre-policy informality rate. Any state-quarter demand shock that affects young cohorts symmetrically cancels in this within-state contrast. Additionally, I augment the panel with state-specific linear time trends so that smooth state-level growth is absorbed. Table A.4 in the Appendix shows that the results are materially unchanged. The stability of the estimated coefficients strongly mitigates concerns for a demand-driven explanation, though non-linear state-specific shocks that interact with both quarter and cohort cannot be fully excluded by design.

Figure 5: Cohort-specific estimates on pre-policy informality rate, $\hat{\beta}_k$



Note: Each point plots the cohort-specific slope of the outcome with respect to the state pre-policy informality rate (2013–2015). Consistent with the exposure schedule, estimates are within a narrow band for unexposed cohorts and shift at the eligibility cutoffs for participation and formality.
Source: Author's calculations with data from ENOE.

Figure 5 plots the cohort-specific estimates on the pre-policy informality rate for the different outcomes of interest. The pattern of estimates is broadly consistent with the exposure to the policy, as defined in figure 1: the gradients sit within a narrow band for cohorts too old to be exposed and move toward zero precisely across the partial-exposure cohorts, most clearly for the participation and formality rates. This graphical evidence points to the relevance of the policy in explaining the results of the long-differencing and pooled regression analyses. The next section develops this cohort-space evidence into formal tests of the identifying assumption.

4.4 Tests to Identifying Assumption

In this section I test the validity of the identifying assumption underlying the previous results. Two related threats could provide alternative explanations for the main estimates. The first is convergence: even absent the policy, participation and formality of younger cohorts in high-informality states might have risen as these poorer, more informal areas catch up with richer, more formal ones. The second is heterogeneous cohort-age profiles: if the age at which young people enter work and formalize varies systematically with a state's informality, cross-cohort gaps would correlate with pre-policy informality mechanically. I address these threats in four steps: formal tests on the cohort-specific gradients, placebo versions of both research designs on untreated cohorts, specifications that allow informality-specific cohort and age profiles, and checks on influential states and on pre-policy trends over time.

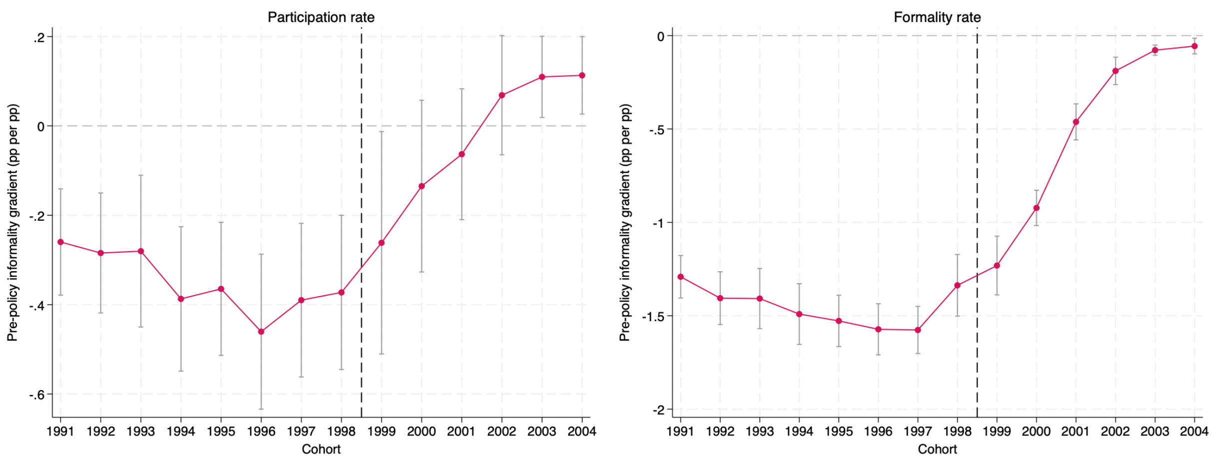
To move from the visual evidence in figure 5 to formal tests, I estimate an alternative, fully interacted version of the cohort-specific equation:

$$Y_{kst} = \alpha_k + \beta_k I_{s,pre} + X_s \Gamma_k + \delta_{kt} + \nu_{kst}$$

in which the intercept, the informality gradient, the control coefficients, and the quarter effects are all allowed to differ by cohort and the equation is estimated jointly on cohort-state-quarter cells over 2016–2019, weighted by the square root of cell population. Joint estimation delivers cohort-specific gradients numerically identical to estimating each cohort separately, but yields a single state-clustered covariance matrix, so hypotheses that compare gradients across cohorts can be tested formally. I focus on participation and formality, the two outcomes for which figure 5 shows systematic cross-cohort patterns. Figure 6 plots the resulting gradients with 95% confidence intervals.

The untreated cohorts' gradients lie in a narrow band (-0.46 to -0.26 for participation, -1.58 to -1.29 for formality) whose width is roughly one fifth of the treatment-induced movement. A formal test rejects their exact equality (participation: $F(7, 31) = 7.4$; formality: $F(7, 31) = 10.8$; both $p < 0.001$), reflecting the precision of the cell-level estimates, but the direction of the variation is the informative object: the estimated linear drift across untreated cohorts is negative for both outcomes (participation: -0.022 per cohort, wild-bootstrap $p = 0.03$; formality: -0.020, $p = 0.15$), opposite in sign to the treatment effect. Smooth cohort trending in high-informality states would push exposed cohorts' gradients further down, not sharply up as observed; under a linear trend counterfactual the baseline estimates are, if anything, conservative. The uptick in the 1998 cohort's gradient relative to 1995–1997 is consistent with partial exposure through the public college channel, which affects the youngest "untreated" cohorts first (39% of the 1995–1998 cohorts are observed in school, either public or private, after the policy).

Figure 6: Cohort-specific pre-policy informality gradient



Note: Each point is the cohort-specific informality gradient β_k from the fully interacted specification described in Section 4.4, in which intercepts, informality gradients, control coefficients, and quarter effects all vary by cohort, estimated jointly on cohort-state-quarter cells over 2016–2019 with square-root-of-cell population weights. Bands are 95% confidence intervals based on standard errors clustered by state. The dashed line separates untreated cohorts (1991–1998) from cohorts with positive exposure.
Source: Author's calculation using data from ENOE.

As a falsification test aimed directly at convergence, I replicate both research designs on cohorts largely untreated by the high-school component of the policy, assigning placebo exposure to the 1995–1998 cohorts with the 1991–1994 cohorts as controls. Although some individuals in the 1995–1998 cohorts may have had partial exposure through public college, they were not subject to the high-school benefit assignment that drives the main analysis. If differential catch-up in high informality states were driving the main estimates, the placebo coefficients should be positive and of comparable magnitude to the baseline. Table 4 shows the opposite for the two key outcomes: for participation and formality, placebo coefficients are statistically indistinguishable from zero or negative, pointing to divergence, with confidence intervals excluding effects as large as the baseline. Panels B and C do show positive and significant placebo estimates for employment and income, but these are not the outcomes driving the paper's main conclusions on formalization and likely reflect underlying age-earnings profiles among these older cohort groups; consistent with this reading, the corresponding long-difference income placebo is large in the uncontrolled specification and vanishes once state controls are included. This pattern undercuts the convergence hypothesis and supports a causal differential ITT interpretation for participation and formality.

Table 4: Placebo Regressions

	(1) Participation rate	(2) Employment rate	(3) Formality rate	(4) Monthly income
Panel A: Long Differences, 2019				
Pre-policy informality rate	-0.11	-0.01	-0.24*	-7.22
	(0.07)	(0.04)	(0.12)	(7.86)
	[0.171]	[0.811]	[0.099]	[0.433]
Full set of controls	✓	✓	✓	✓
Mean dependent variable	-12.47	-2.17	-8.07	-1,050.38
Observations	128	128	128	128
Panel B: Pooled Regressions, 2016–2019				
Pre-policy informality rate × Exposure	-0.09**	0.04*	-0.10	10.83**
	(0.03)	(0.02)	(0.05)	(3.23)
Full set of controls	✓	✓	✓	✓
Mean dependent variable	60.42	93.00	40.45	5,288.04
Observations	4,096	4,096	4,096	4,096
Panel C: Pooled Regressions, 2016				
Pre-policy informality rate × Exposure	-0.01	0.08**	0.14	10.11**
	(0.04)	(0.03)	(0.09)	(2.94)
Full set of controls	✓	✓	✓	✓
Mean dependent variable	54.39	91.88	34.81	4,490.38
Observations	1,024	1,024	1,024	1,024

Note: Placebo exposed cohorts are 1995–1998, with 1991–1994 as controls. Standard errors clustered by states in parentheses, wild cluster bootstrap p-values for the pre-policy informality coefficient in brackets for the long differences. The square root of cell sizes is used to construct weights for the observations. Cells are weighted by the square root of the cell population (the square root of the positive-income cell population in column (4), which restricts to individuals reporting strictly positive income).
***p<0.01, **p<0.05, *p<0.1

The placebo evidence rules out catch-up among older cohorts, but heterogeneous profiles could in principle operate smoothly across all cohorts. Table A.5 therefore augments the panel specification with interactions of pre-policy informality and smooth cohort or age profiles, extending the sample to cohorts 1991–2004 so that the added untreated cohorts help pin down the profiles. The formality effect is robust throughout, ranging from 0.78 to 1.21 and always significant at the 1 percent level by wild cluster bootstrap. The participation effect attenuates to 0.15–0.22 but remains positive and significant at conventional levels. Given the exposure function ramps from zero to one over only four cohorts, flexible informality-specific trends absorb part of the exposure variation by construction, so these estimates should be read as conservative bounds rather than preferred point estimates.

The estimates are also not driven by influential states. Re-estimating the main specifications excluding one state at a time yields formality coefficients ranging from 0.79 to 0.98 in the long differences and from 1.24 to 1.31 in the panel, always significant at the 1 percent level; participation coefficients are similarly stable. In particular, excluding jointly the three southern states with the highest informality and the most atypical age-participation profiles (Chiapas, Guerrero and Oaxaca) leaves the estimates essentially unchanged (Appendix Table A.3 reports the full set of inference and leverage checks).

Finally, I estimate pre-policy trends over time. Table A.6 conditions the long differences on each state's cross-cohort gaps measured three years before the start of the policy. The participation estimate is unchanged (0.63 versus 0.59 at baseline), and the formality estimate remains large and significant at the 5 percent level (0.58, wild-bootstrap $p = 0.02$), about two thirds of the baseline. The lagged formality gap itself enters positively, indicating that part of the raw formality gradient reflects persistent state-level differences in young-versus-old formality gaps. It is reassuring that the estimate net of this persistence remains close to the profile-augmented panel estimates of Table A.5, as two independent adjustments (conditioning on lagged gaps in the cross-section and allowing informality-specific cohort and age profiles in the panel) bound the formality effect within the same range. Figure A.3 in the Appendix provides the corresponding visual evidence: quarterly cohort gaps between the treated (15–18) and control (21–24) age groups, computed by state since 2010, smoothed by locally weighted regression, and split at the median pre-policy informality rate, evolve in parallel across the two groups of states before 2016 and diverge only after policy implementation.

5 School Enrollment and Robustness

An important consideration is whether this policy had an impact on the schooling decisions of individuals. The new medical benefits associated with being enrolled in a public high school or college can increase school enrollment and attainment to the detriment of participation in the labor market at younger ages, which could be a desirable outcome from both the perspective of the individual and society as a whole. The results of the previous section of an increased participation rate among exposed cohorts in high informal states suggest that this is not the case. In this section, I explore more thoroughly the impact on school enrollment as well as differentiate the main results by school enrollment to better understand the mechanisms at play. I then look at the heterogeneous results and robustness to excluding the self-employed.

5.1 School enrollment

First, I explore the impact of the social insurance expansion policy on school enrollment rates of the cohorts analyzed. Table 5 shows estimates of the long differences and panel equations presented in sections 4.3.1 and 4.3.2, using school enrollment rates as the dependent variable. There is no evidence of an impact of the policy on schooling rates: neither the long-difference estimate nor the panel estimate is statistically significant, and neither is economically large.

Table 5: Long Differences and Panel Estimates of School Enrollment Rates

	(1) Long Differences	(2) Panel Estimates
Pre-policy informality rate	-0.21	-0.12
	(0.15)	(0.09)
	[0.220]	[0.390]
Log GDP per capita in 2013	-9.36***	-1.76
	(3.27)	(1.60)
Educational attainment in 2015	4.41*	-2.55*
	(2.28)	(1.38)
Regional controls	✓	
Cohort FE		✓
State FE		✓
Quarter FE	✓	✓
Mean dependent variable	46.42	63.97
Observations	128	5,120

Note: Standard errors clustered by state in parentheses; wild cluster bootstrap p-values for the pre-policy informality coefficient in brackets (Webb weights, 9,999 replications). The square root of cell population is used as weights for the observations. For column (2) the independent variables are interacted with the policy exposure function as in Table 3.
***p<0.01, **p<0.05, *p<0.1

Now, I distinguish the main results of the previous section between workers going to school at the time of the survey and those not going. Table 6 presents the estimates of long differences on the main outcomes of the labor market distinguishing by school enrollment. Panel A shows the main results, considering the full sample, which are exactly those presented previously in Table 2. Panel B shows the long differences estimates using the sample of people not going to school and Panel C shows the results using the sample of students at the time of the survey. Approximately 53% of the full sample studied is enrolled in school. The results for participation rate, formality rate, and monthly income are higher among the sample that is going to school. In the case of the formality rate, one percentage point increase in the pre-policy informality rate is associated with an increase of 0.95 percentage points in the formality rate of the cohorts affected by the policy. Formality is significant at the 1 percent level by wild cluster bootstrap in both subsamples, and participation in both as well (at the 5 percent level among those not in school); the smaller employment and income estimates are at most marginally significant.

To see this graphically, figure 7 shows the estimated coefficients of the pre-policy informality rate on the main outcome variables distinguishing by school enrollment. This figure presents the results from both the long differences and panel specifications. For the case of the panel specification the coefficient corresponds to the interaction of the pre-policy informality rate and the exposure function. We confirm that the estimated coefficients are higher for the subsample of people that are in school, across the specifications and main outcomes (with the exception of the employment rate).

An important caveat is that school enrollment in Table 6 is measured contemporaneously with labor market outcomes (that is, after policy exposure) so the in-school and not-in-school subgroups are not fixed populations. Even though Table 5 finds no clear average enrollment response, conditioning on current enrollment may still shift subgroup composition in ways that vary with pre-policy informality. The in-school and not-in-school estimates should therefore be read as suggestive heterogeneity rather than clean effects for fixed student and non-student populations. With that caveat in mind, the results are consistent across both subsamples. In particular, they provide suggestive evidence that students affected by the social insurance expansion policy are not more likely to be employed informally, despite the reduction in the relative cost of an informal job while also being enrolled in school. This is more relevant as there is evidence that there was an increase in the participation rate among exposed cohorts in states with higher informality.

Table 6: Cross Cohort Long Differences by School Enrollment, 2019

	(1) Participation rate	(2) Employment rate	(3) Formality rate	(4) Monthly income
Panel A: Full Sample				
Pre-policy informality rate	0.59***	0.15*	0.87***	11.18
	(0.09)	(0.06)	(0.16)	(10.00)
	[<0.001]	[0.053]	[0.003]	[0.278]
Log GDP per capita in 2013	6.04***	1.58	1.91	-134.53
	(0.92)	(0.99)	(1.33)	(128.86)
Educational attainment in 2015	-3.29***	-0.37	2.25	-328.57**
	(1.10)	(0.82)	(1.88)	(137.75)
Regional controls	✓	✓	✓	✓
Quarter FE	✓	✓	✓	✓
Mean dependent variable	-37.89	0.08	-32.75	-1,930.18
Panel B: Sample Not In School				
Pre-policy informality rate	0.37**	0.10	0.83***	18.86
	(0.13)	(0.07)	(0.17)	(12.54)
	[0.039]	[0.255]	[0.007]	[0.155]
Log GDP per capita in 2013	-1.02	2.43**	2.32	-247.01*
	(3.49)	(1.11)	(1.58)	(135.31)
Educational attainment in 2015	-2.65	-1.80	2.59	-110.39
	(2.17)	(1.07)	(1.86)	(175.33)
Regional controls	✓	✓	✓	✓
Quarter FE	✓	✓	✓	✓
Mean dependent variable	-17.25	-1.13	-31.43	-1,493.67
Panel C: Sample In School				
Pre-policy informality rate	0.56***	0.20*	0.95***	29.16
	(0.08)	(0.09)	(0.19)	(15.33)
	[<0.001]	[0.076]	[0.001]	[0.115]
Log GDP per capita in 2013	4.06*	1.20	-2.91	-363.24
	(2.13)	(1.33)	(4.34)	(494.58)
Educational attainment in 2015	-0.79	1.52	5.07	49.21
	(1.70)	(0.92)	(3.80)	(269.29)
Regional controls	✓	✓	✓	✓
Quarter FE	✓	✓	✓	✓
Mean dependent variable	-21.95	0.68	-30.78	-2,120.29
Observations	128	128	128	128

Note: Standard errors clustered by state in parentheses; wild cluster bootstrap p-values for the pre-policy informality coefficient in brackets (Webb weights, 9,999 replications); stars on that coefficient are based on the wild-bootstrap p-values. Panels B and C restrict the sample to individuals not attending and attending school at the time of the survey, respectively, with cell rates computed over each subsample's population. Cells are weighted by the square root of the cell population (the square root of the positive-income cell population in column (4), which restricts to individuals reporting strictly positive income). Regional controls and quarter fixed effects included in all panels.
***p<0.01, **p<0.05, *p<0.1

Figure 7: Estimates on pre-policy informality rate by school enrollment

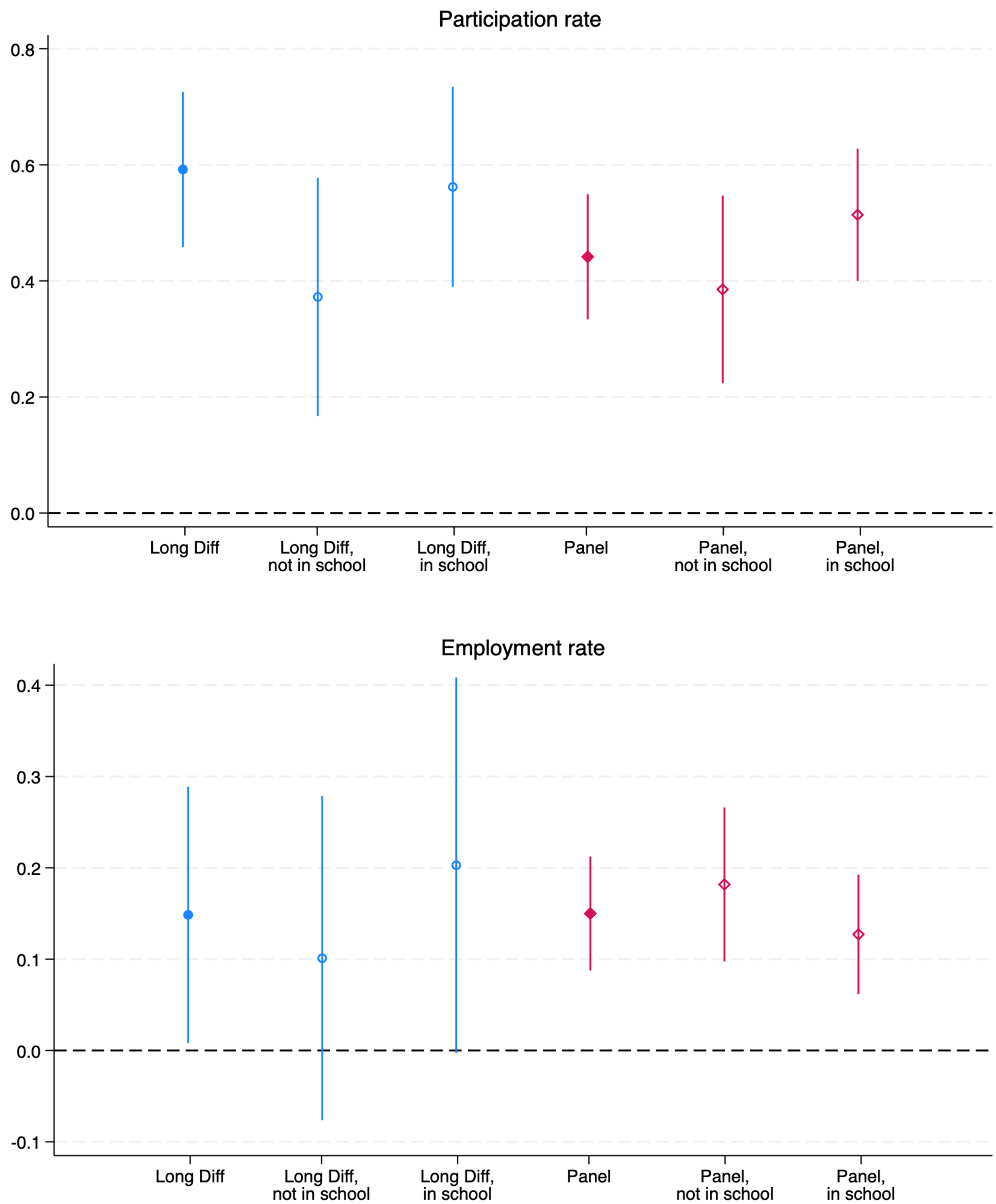
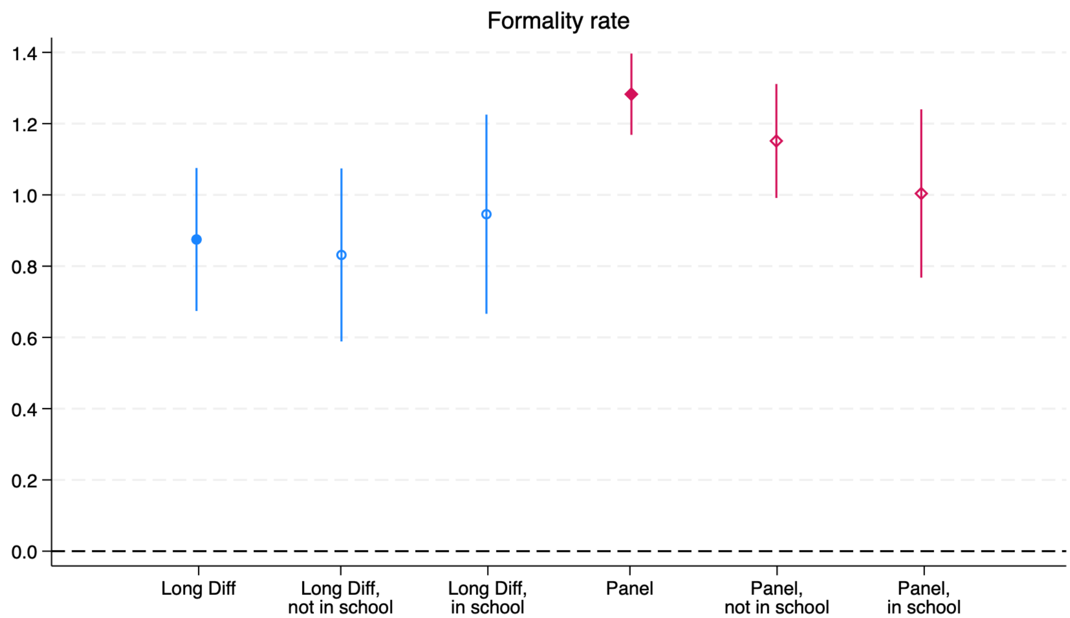
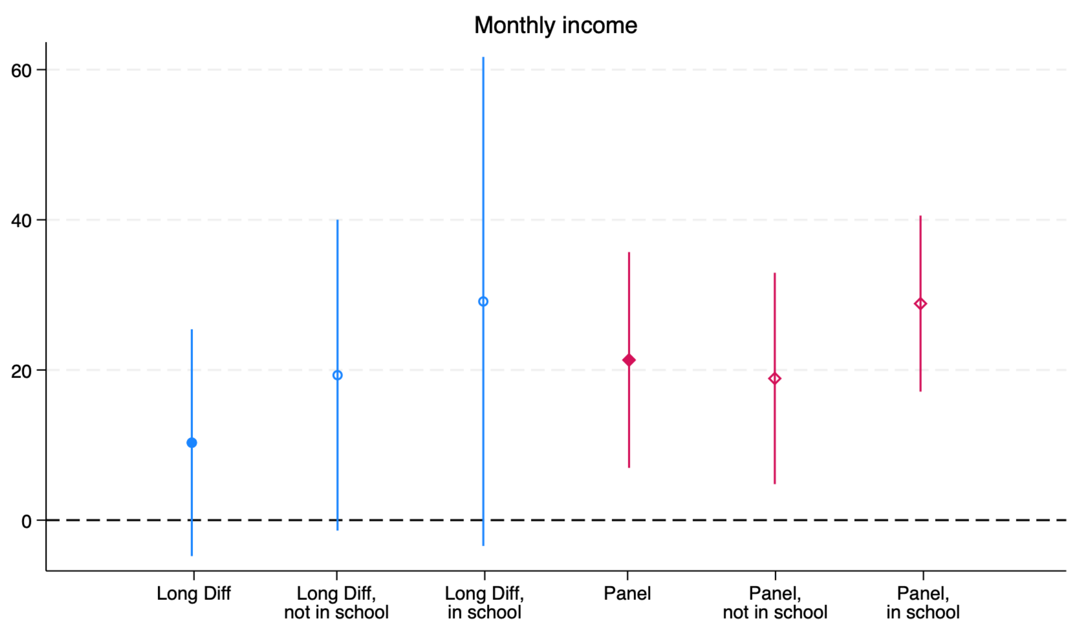


Figure 7: Estimates on pre-policy informality rate by school enrollment (cont.)





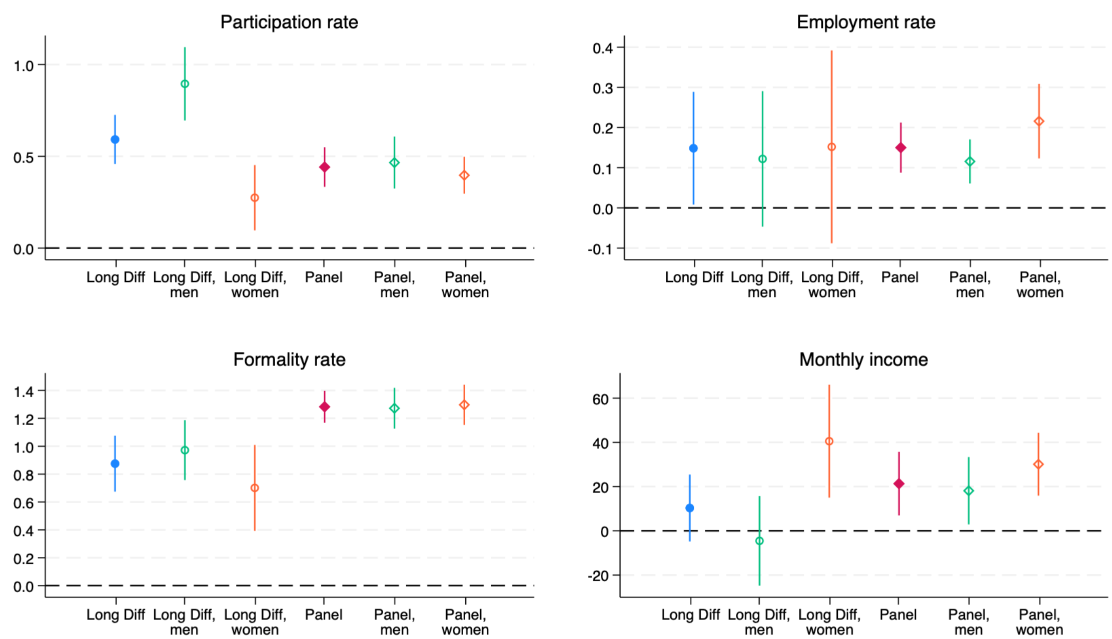
Note: Coefficients on the pre-policy informality rate are plotted, separately for those in school and not in school. Long difference estimates (2019) use cohort gaps regressed on state informality (2013–2015). Panel estimates (2016–2019) report the coefficient on the interaction between state informality (2013–2015) and the policy exposure function by cohort.
Source: Author's calculations with data from ENOE.

It is uncertain whether it is desirable to have more individuals start working while they study. Employment spells while studying could provide valuable training experiences. On the other hand, it could deter from further investing in schooling (Atkin, 2016). However, if anything, it is desirable to have students work formally vs. informally at the beginning of their labor market trajectories as there are negative employment and earning effects associated with youth informality spells.

5.2 Heterogeneous Results

I distinguish the main results among the sample of men and women to explore any differential impact of the policy by gender. The sample is roughly evenly split between men and women. Figure 8 presents the graphical results of the coefficient on the pre-policy informality rate between specifications and by gender.

Figure 8: Estimates on pre-policy informality rate by gender

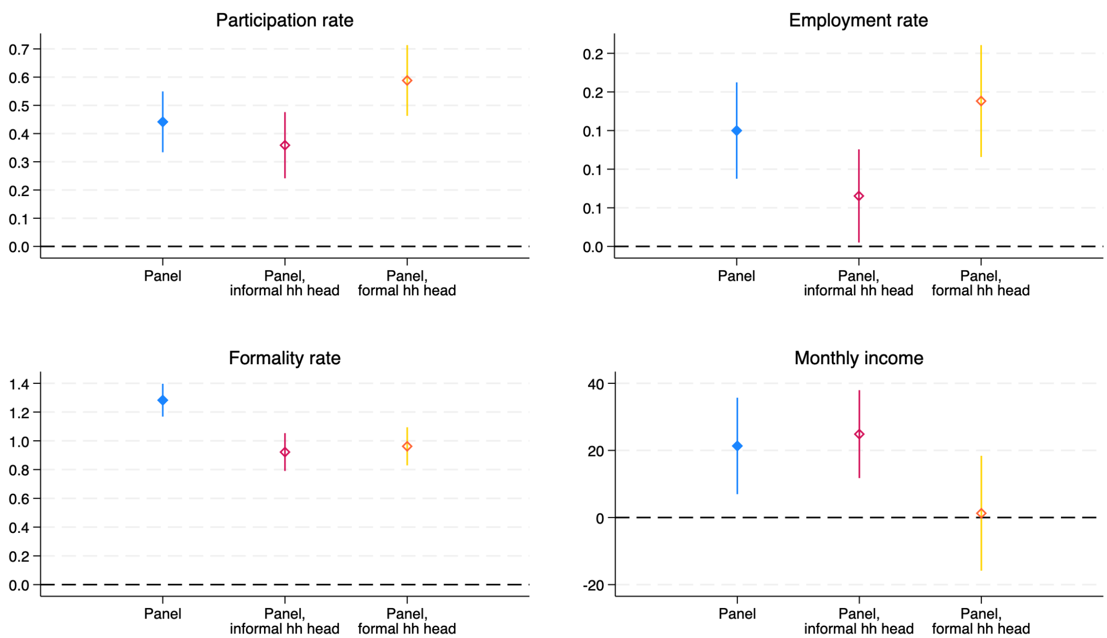


Note: Coefficients on the pre-policy informality rate are plotted, separately for men and women. Long difference estimates (2019) use cohort gaps regressed on state informality (2013-2015). Panel estimates (2016–2019) report the coefficient on the interaction between state informality (2013–2015) and the policy exposure function by cohort.
Source: Author's calculations with data from ENOE.

From these results, we cannot observe a stark difference in the impact of the policy between men and women. However, it is the case that the point estimates for the participation and formality rate are higher for men than for women, whereas the point estimates for the employment rate and monthly income are higher for women. Overall, I do not find significant differences in the impact of the policy by gender.

I further make a distinction by the formality of the head of household. Figure 9 shows that while the impact on the participation rate is larger for individuals with a formal head of household, the formalization effect is similar regardless. The fact that the results are not local to fully informal households highlights the importance of having access to social insurance benefits directly, not only through a parent.

Figure 9: Estimates on pre-policy informality rate by formality of head of household (hh)



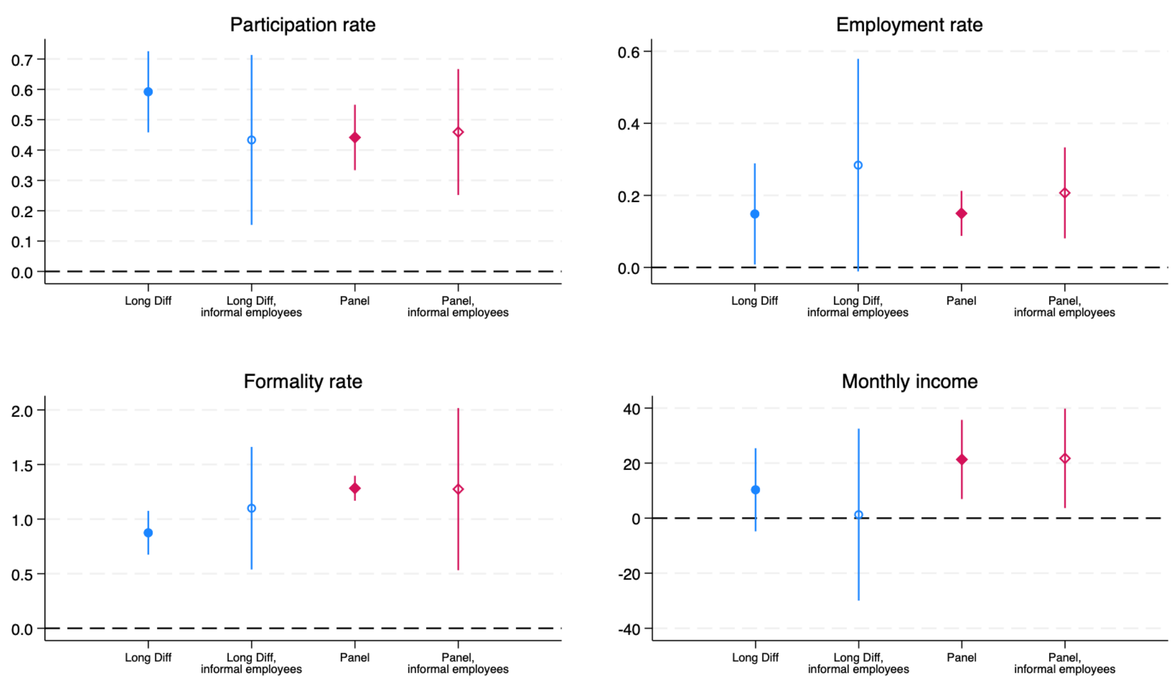
Note: Coefficients on the pre-policy informality rate are plotted, separately for those individuals with formal and informal household head. Long difference estimates (2019) use cohort gaps regressed on state informality (2013–2015). Panel estimates (2016–2019) report the coefficient on the interaction between state informality (2013–2015) and the policy exposure function by cohort.
Source: Author's calculations with data from ENOE.

5.3 Alternative Sample of Informal Workers

Up to this point, I have been using the official definition of informal employment established by the Mexican National Institute of Statistics and Geography (INEGI). This definition includes both employees and self-employed individuals. I explore how the main results change if we focus only on subordinate employment, i.e., employees without access to social insurance. I use this alternative sample that only considers employees for the pre-policy period as well as for the main period of analysis (see figure A.4 for the spatial distribution of the alternative pre-policy informality across states).

Figure 10 presents the main results of the long differences and the panel specifications with full controls along with the results using the alternative sample of workers, considering only employees. As expected, the alternative estimates on the pre-policy informality rate become noisier but are similar in magnitude to the main estimates. The panel estimates suggest a positive and significant impact of the policy on the participation and formality rates among exposed cohorts in states with high levels of informality.

Figure 10: Estimates on pre-policy informality rate, alternative definitions of informality



Note: Coefficients on the pre-policy informality rate are plotted, separately for the full sample and an alternative sample that considers only employees. Long difference estimates (2019) use cohort gaps regressed on state informality (2013–2015). Panel estimates (2016–2019) report the coefficient on the interaction between state informality (2013–2015) and the policy exposure function by cohort.
Source: Author's calculations with data from ENOE.

5.4 Measurement of Exposure

The baseline design assigns exposure at the cohort level, but effective exposure to the school-linked benefit required being enrolled, and enrolled in a public institution, both of which vary across states. To assess whether this measurement matters, I construct a predicted exposure measure that scales cohort exposure by each state's pre-policy school-attendance rate of 15–18 year-olds (ENOE, 2013–2015; mean 71.5% ranging from 58% to 83% across states) and by the public share of upper-secondary enrollment in the 2015–2016 school year (roughly 81% nationally, see figure A.5). Table A.7 re-estimates the panel specification replacing cohort exposure with predicted exposure. The cross-state pattern is unchanged in sign and significance, and the coefficients rise as expected when the treatment variable is scaled toward actual eligibility (for formality, from 1.28 at baseline to 1.59 per unit of predicted exposure, with similar results for participation) indicating that the findings are not an artifact of assuming uniform exposure within cohorts.

The same ingredients discipline the interpretation of magnitudes and of the contamination discussed in the research design. Two facts stand out. First, partial treatment of the comparison group is substantial: 39% of the 1995–1998 cohorts are observed attending school during 2016–2019. Second, neither ingredient is orthogonal to initial informality: the pre-policy enrollment rate and the college contamination rate both decline significantly with pre-policy informality (Table A.7; figure A.6). On net, these correlations work in a conservative direction. Lower enrollment in high informality states means the intention-to-treat estimates understate per-student effects precisely where the identifying variation is largest, and although lower contamination in high informality states works in the opposite direction, the level of contamination dominates its gradient in every state, so cohort contrasts remain attenuated throughout (footnote 8 states the exact condition).

Finally, these ingredients translate the intention-to-treat estimates into magnitudes per enrolled-eligible student. The exposure differential between designated exposed and unexposed cohorts is at most 0.58 (the product of the enrollment rate and the public share) and smaller once the comparison group's partial treatment is counted. Dividing by this differential implies effects per enrolled-eligible student of at least 1.7 times the baseline estimates, and larger under fuller accounting of contamination.

6 Labor Market Insertion

I complement my main results with information from ENILEMS, a specialized survey of 18-20 year old high-school graduates. This dataset offers richer detail on students' high-school histories and subsequent labor market experiences. Crucially, it distinguishes between public and private high schools, facilitating a cleaner identification of individuals' policy exposure, and it is fully comparable with ENOE. I use the 2016 and 2019 waves of the survey, which allows me to compare cohorts exposed to, and unaffected by the policy. ENILEMS is conducted in the third quarter of the year, after the school year has finalized. Therefore, the cohorts surveyed in 2016 were not exposed to the policy (or minimally during their last year of high school), and the cohorts surveyed in 2019 were partially and fully exposed to the policy. Specifically, the two waves of ENILEMS contain information for cohorts born between 1996 and 2001.

Table 7: Labor Market Insertion Data (ENILEMS) - Summary Statistics

	(1) 2016 survey	(2) 2019 survey	(3) Difference
Female	0.54	0.53	-0.00 (-0.14)
Public high school	0.83	0.83	0.01 (0.71)
Had job during high school	0.16	0.20	0.04*** (4.61)
Continues in first job	0.35	0.41	0.06*** (3.96)
Participation rate	0.42	0.56	0.13*** (10.32)
Employment rate	0.89	0.92	0.03*** (2.76)
Formality rate	0.34	0.32	-0.02 (-1.09)
Real monthly income	3,066	3,765	699*** (6.25)
Attempted higher education	0.40	0.45	0.05*** (4.18)
Ever enrolled in higher education	0.26	0.34	0.08*** (6.77)
Either parent with higher education	0.21	0.23	0.02 (1.45)
Observations	7,191	7,799	

Note: t-statistic in parentheses. Survey weights used.
***p<0.01, **p<0.05, *p<0.1

Table 7 presents summary statistics for both waves of the survey. Each wave contains information for more than 7 thousand individuals, representative of a population of between 2.1 and 2.9 million. Around 83% of high-school graduates were enrolled in a public school across both waves. There were some significant changes in the labor market outcomes of high-school graduates between both surveys. There was a significant increase in the share of individuals that worked during high school and those that continued working in their first job. We observe a sizable increase in the participation rate (from 42% to 56%). Additionally, there were also increases in the employment rate and the real monthly income. Along schooling outcomes there were significant changes as well, with an increase of the share of high-school graduates that attempted and were ever enrolled in a higher education institution.

I estimate panel specifications similar to those presented in section 4.3.2 and Table 3. I take advantage of the information of type of high school (public vs. private) to refine my exposure policy function. I only considered as exposed those cohorts that were born after 1998 and that went to a public high school. To account for labor market and family background conditions across cohorts, states, and waves, I interact exposure with state GDP (to control for differential economic conditions), the share with at least one parent with higher education, and the share that worked during high school. Note that unlike standard additive controls, these variables enter as interactions with exposure, so they capture both baseline heterogeneity and how the policy's effect varies along each dimension; the high-school work share interaction is interpreted substantively below.

Table 8 presents the main results of the panel estimates using the labor market insertion data from ENILEMS. The interaction of the refined exposure policy and the pre-policy informality rate is associated with a significant increase in the participation rate (at the 5% level) and a marginally significant increase in the formality rate (coefficient 1.27, SE 0.70, $t \approx 1.81$). The estimate for the employment rate is not significant. These magnitudes are broadly consistent with the main results in Table 3.

Two further patterns emerge from the high-school work share interaction. First, among exposed cohorts, states where a larger share of students worked during high school exhibit a *lower* formality rate, conditional on the pre-policy informality level. This is consistent with the benefit-substitution channel: students already accustomed to combining work and school may have reduced their relative valuation of formal employment while benefits were available through school. This points to the substitution margin operating for the study-and-work group. It does not, however, overturn the aggregate pattern: informal employment declines as a share of the cohort population precisely where exposure was strongest (Table A.2), so benefit substitution appears to operate at the margin for this group without producing a net shift toward informality. Second, those same states show a *higher* share of higher-education enrollment, suggesting the policy also raised the perceived value of continuing in the public education system for that group. Note that the pre-policy informality interaction does not drive either of these patterns (its coefficient is positive but imprecise for formality, and negative and imprecise for higher-education enrollment) so neither finding should be read as a comparison between more- and less-informal states.

Table 8: Labor Market Insertion Panel Estimates, 2016–2019

	(1) Participation rate	(2) Employment rate	(3) Formality rate	(4) Monthly income	(5) Ever enrolled in higher ed
Pre-policy informality	1.31**	0.15	1.27*	103.52***	-1.17
rate × Exposure	(0.57)	(0.28)	(0.70)	(33.84)	(1.04)
Log state GDP	-0.10**	-0.02	-0.08	-6.25***	0.10
× Exposure	(0.04)	(0.02)	(0.05)	(1.97)	(0.08)
Share with higher ed	-0.53	0.46	0.02	-53.73*	-0.55
parents × Exposure	(0.40)	(0.31)	(0.37)	(30.24)	(0.72)
Share that worked	0.12	-0.02	-1.06***	-31.39	1.38***
in HS × Exposure	(0.25)	(0.14)	(0.27)	(22.76)	(0.47)
Public/Private HS FE	✓	✓	✓	✓	✓
Cohort FE	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓	✓
Survey wave FE	✓	✓	✓	✓	✓
Mean dependent variable	50.17	90.70	33.89	4,212.60	31.15
Observations	376	373	319	372	374

Note: Standard errors clustered by state in parentheses. The square root of cell sizes are used as weights for the observations. Cell sizes are identical across specifications, except for the income one as we restrict to individuals reporting strictly positive income.
***p<0.01, **p<0.05, *p<0.1

7 Conclusion

Informal employment remains a defining feature of labor markets in developing countries and is especially prevalent at labor market entry. Because early informal spells can scar future formal attachment and earnings, understanding whether school-linked access to social insurance alters early choices is policy-relevant.

This paper exploits a cohort-by-state design to study Mexico's 2016 policy that made public high-school students eligible for IMSS medical benefits while enrolled. Comparing exposed and slightly older adjacent cohorts and allowing effects to vary with pre-policy state informality, I estimate a differential intention-to-treat effect: the policy's impact is larger in initially more informal states. Exposed cohorts in high-informality states exhibit higher participation and markedly higher formal employment relative to adjacent unexposed cohorts, compared with the same cohort gap in low-informality states. Point estimates imply that a 1-percentage-point increase in initial informality is associated with a 0.8–1.3-percentage-point wider exposed-minus-unexposed cohort gap in the formality rate. To gauge magnitudes, the standard deviation of pre-policy informality across states is 13.1 percentage points, so a one-standard-deviation shift in initial informality corresponds to an 11.4–16.7-percentage-point change in the cohort formality gap.[11] Against the average gap of −32.8 percentage points (exposed cohorts are young and far less formal than their older neighbors everywhere) this amounts to closing roughly one third to one half of the cross-cohort formality gap, from a low exposed-cohort base (a formality rate of 7.6 percent, and a formal-employment-to-population ratio of 1.3 percent, in 2019).

The population-based decomposition (Table A.2) shows how this occurs: the participation response flows entirely into employment, formal employment absorbs more than the full increase, informal employment declines, and unemployment is unchanged. The evidence points to formal sector entry rather than a shift toward informality. I find no effects on school enrollment and only very small changes in income. Effects are similar for women and men, for students and non-students, and robust to excluding the self-employed. Participation estimates range from 0.15 to 0.59 depending on how flexibly cohort profiles are allowed to vary with initial informality, remaining positive and significant throughout. Formality estimates remain large (between 0.78 and 1.21 under the same variations) and significant at the 1 percent level in every specification.

Taken together, the evidence is most consistent with a salience and administrative-frictions channel operating at the school-to-work transition: benefits become concrete, and a permanent social security number lowers hiring frictions. Consistent with a channel that works through formal hiring by firms, the effects are concentrated in subordinate wage employment, with no response of self-employment on either the formal or informal margin. I find no evidence that delinked benefits produced a net shift toward informality

[11] The population-weighted standard deviation of pre-policy informality is 10.6 percentage points, which gives a scaled range of 9.2–13.6 percentage points.

(informal employment declines precisely where exposure was strongest) although the heterogeneity among graduates who worked during high school suggests the substitution margin may operate for that group. The results may be read as lower bounds on the schooling-linked impact under the assumption that public college eligibility generates effects in the same direction as high-school eligibility: 39% of the comparison cohorts are observed attending school (either public or private) after the policy, which attenuates the exposed-minus-unexposed gap toward zero; this attenuation dominates its covariation with initial informality in every state. If college effects operate differently, the lower-bound interpretation need not hold. Scaling by measured enrollment and public-school shares, the estimates imply effects per enrolled-eligible student of at least 1.7 times the intention-to-treat magnitudes. The estimates also pertain to the early-career margin and a pre-pandemic window (2016–2019).

For policy design, these findings suggest that school-linked health coverage can complement formal labor-market entry without inducing a shift toward informality, particularly where informality is initially high. A full welfare evaluation would integrate labor market effects with health benefits and longer-run outcomes; the college component and post-2019 dynamics are fruitful next steps. More broadly, advancing formalization will require understanding both workers' early-career decisions and firms' hiring margins in high-informality environments. Finally, given the relative stability of informality rates in countries such as Mexico, it is relevant to assess which policies foster formalization and which hinder it. Improving the working conditions of millions of people in the developing world remains one of the main challenges from a policy perspective.

References

Akay, A. and Khamis, M. (2012). The Persistence of Informality: Evidence from Panel Data. *Informal Employment in Emerging and Transition Economies (Research in Labor Economics, Volume 34). Emerald Group Publishing Limited*, pages 229–255.

Akosa Antwi, Y., Moriya, A. S., and Simon, K. (2013). Effects of Federal Policy to Insure Young Adults: Evidence from the 2010 Affordable Care Act's Dependent-Coverage Mandate. *American Economic Journal: Economic Policy*, 5(4):1–28.

Alonso-Ortiz, J. and Leal, J. (2018). Cross-Subsidies, and the Elasticity of Informality to Social Expenditures: The Case of Mexico's Seguro Popular. *Review of Income and Wealth*, 64(2):482–512.

Alvarez, J. and Pizzinelli, C. (2021). COVID-19 and the Informality-driven Recovery: The Case of Colombia's Labor Market. *IMF Working Paper*.

Alzúa, M. L., Cruces, G., and Ripani, L. (2013). Welfare Programs and Labor Supply in Developing Countries: Experimental Evidence from Latin America. *Journal of Population Economics*, 26(4):1255–1284.

Arias, J., Azuara, O., Bernal, P., Heckman, J. J., and Villarreal, C. (2010). Policies to Promote Growth and Economic Efficiency in Mexico. Technical report, National Bureau of Economic Research.

Arulampalam, W. (2001). Is Unemployment Really Scarring? Effects of Unemployment Experiences on Wages. *The Economic Journal*, 111(475):F585–F606.

Ashworth, J., Hotz, V. J., Maurel, A., and Ransom, T. (2021). Changes across Cohorts in Wage Returns to Schooling and Early Work Experiences. *Journal of Labor Economics*, 39(4):931–964.

Atkin, D. (2016). Endogenous Skill Acquisition and Export Manufacturing in Mexico. *American Economic Review*, 106(8):2046–85.

Azuara, O. and Marinescu, I. (2013). Informality and the Expansion of Social Protection Programs: Evidence from Mexico. *Journal of Health Economics*, 32(5):938–950.

Bachas, P., Gadenne, L., and Jensen, A. (2023). Informality, Consumption Taxes, and Redistribution. *The Review of Economic Studies*, 91(5):2604–2634.

Bell, D. N. and Blanchflower, D. G. (2011). Young People and the Great Recession. *Oxford Review of Economic Policy*, 27(2):241–267.

Berniell, I., Gasparini, L., Marchionni, M., and Viollaz, M. (2023). Lucky Women in Unlucky Cohorts: Gender Differences in the Effects of Initial Labor Market Conditions in Latin America. *Journal of Development Economics*, 161:103042.

Bleakley, H. (2010). Malaria Eradication in the Americas: A Retrospective Analysis of Childhood Exposure. *American Economic Journal: Applied Economics*, 2(2):1–45.

Bosch, M. and Campos-Vazquez, R. M. (2014). The Trade-Offs of Welfare Policies in Labor Markets with Informal Jobs: The Case of the "Seguro Popular" Program in Mexico. *American Economic Journal: Economic Policy*, 6(4):71–99.

Bosch, M. and Maloney, W. F. (2010). Comparative Analysis of Labor Market Dynamics Using Markov Processes: An Application to Informality. *Labour Economics*, 17(4):621–631.

Breza, E. and Kaur, S. (2025). Labor Markets in Developing Countries. *Annual Review of Economics*, 17(Volume 17, 2025):747–776.

Brockmeyer, A., Gerard, F., Ulyssea, G., and Wu, L. (2025). Formal Labor Market Dynamics in Developing Countries. *Working Paper*.

Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427.

Campos-Vázquez, R. M. and Knox, M. A. (2013). Social Protection Programs and Employment: The Case of Mexico's Seguro Popular Program. *Economía Mexicana. Nueva Época*, 22(2):403–448.

Cruces, G., Ham, A., and Viollaz, M. (2012). Scarring Effects of Youth Unemployment and Informality: Evidence from Argentina and Brazil. *Working Paper*.

Donovan, K., Lu, W. J., and Schoellman, T. (2023). Labor Market Dynamics and Development. *The Quarterly Journal of Economics*, 138(4):2287–2325.

Duflo, E. (2001). Schooling and Labor Market Consequences of School Construction in Indonesia: Evidence from an Unusual Policy Experiment. *American Economic Review*, 91(4):795–813.

Esquivel, G. and Ordaz-Díaz, J. L. (2009). *¿Es Correcto Vincular la Política Social a la Informalidad en México?: Una Prueba Simple de las Premisas de esta Hipótesis*. CEPAL.

Gerard, F., Naritomi, J., and Silva, J. (2021). Cash Transfers and Formal Labor Markets: Evidence from Brazil. *CEPR Discussion Paper*.

Guner, N., Kulikova, Y. A., and Valladares-Esteban, A. (2025). Does the Added Worker Effect Matter? *Review of Economic Dynamics*, 56:101271.

Harris, J. R. and Todaro, M. P. (1970). Migration, Unemployment and Development: A Two-Sector Analysis. *The American Economic Review*, 60(1):126–142.

Hotz, V. J., Xu, L. C., Tienda, M., and Ahituv, A. (2002). Are There Returns to the Wages of Young Men from Working While in School? *The Review of Economics and Statistics*, 84(2):221–236.

ILO (2018). Women and Men in the Informal Economy: A Statistical Picture. Technical Report 46, International Labour Organization.

Jensen, A. (2022). Employment Structure and the Rise of the Modern Tax System. *American Economic Review*, 112(1):213–34.

Le Barbanchon, T., Ubfal, D., and Araya, F. (2023). The Effects of Working While in School: Evidence from Employment Lotteries. *American Economic Journal: Applied Economics*, 15(1):383–410.

Levy, S. (2007). ¿Pueden los Programas Sociales Disminuir la Productividad y el Crecimiento Económico? Una Hipótesis para México. *El Trimestre Económico*, 74(295):491–540.

Levy, S. (2008). *Good Intentions, Bad Outcomes: Social Policy, Informality, and Economic Growth in Mexico*. Rowman & Littlefield.

Leyva, G. and Urrutia, C. (2023). Informal Labor Markets in Times of Pandemic. *Review of Economic Dynamics*, 47:158–185.

MacKinnon, J. G., Ørregaard Nielsen, M., and Webb, M. D. (2023). Cluster-robust inference: A guide to empirical practice. *Journal of Econometrics*, 232(2):272–299.

Maloney, W. F. (2004). Informality Revisited. *World Development*, 32(7):1159–1178.

Meghir, C. and Phillips, D. (2008). Labour Supply and Taxation. Technical report, IFS Working Papers.

Moffitt, R. A. (2002). Welfare Programs and Labor Supply. *Handbook of Public Economics*, 4:2393–2430.

Osuna-Gómez, D. (2023). The Impact of the COVID-19 Pandemic on Post-Great Recession Entrants: Evidence from Mexico. *Labour Economics*, 81:102337.

Portes, A., Castells, M., and Benton, L. (1989). *The Informal Economy: Studies in Advanced and Less Developed Countries*. A Johns Hopkins paperback. Johns Hopkins University Press.

Roodman, D., Nielsen, M. Ø., MacKinnon, J. G., and Webb, M. D. (2019). Fast and wild: Bootstrap inference in stata using boottest. *The Stata Journal*, 19(1):4–60.

Ruhm, C. J. (1997). Is High School Employment Consumption or Investment? *Journal of Labor Economics*, 15(4):735–776.

Samaniego de la Parra, B. and Fernández Bujanda, L. (2024). Increasing the Cost of Informal Employment: Evidence from Mexico. *American Economic Journal: Applied Economics*, 16(1):377–411.

Schwandt, H. and von Wachter, T. (2019). Unlucky Cohorts: Estimating the Long-Term Effects of Entering the Labor Market in a Recession in Large Cross-Sectional Data Sets. *Journal of Labor Economics*, 37(S1):S161–S198.

Seira, E., Meza, I., González-Pier, E., and Alcaraz Prous, E. (2023). Did Mexico's Seguro Popular Universal Health Coverage Programme Really Reduce Formal Jobs? Technical Report 46, WIEGO.

Solon, G., Haider, S. J., and Wooldridge, J. M. (2015). What are we weighting for? *The Journal of Human Resources*, 50(2):301–316.

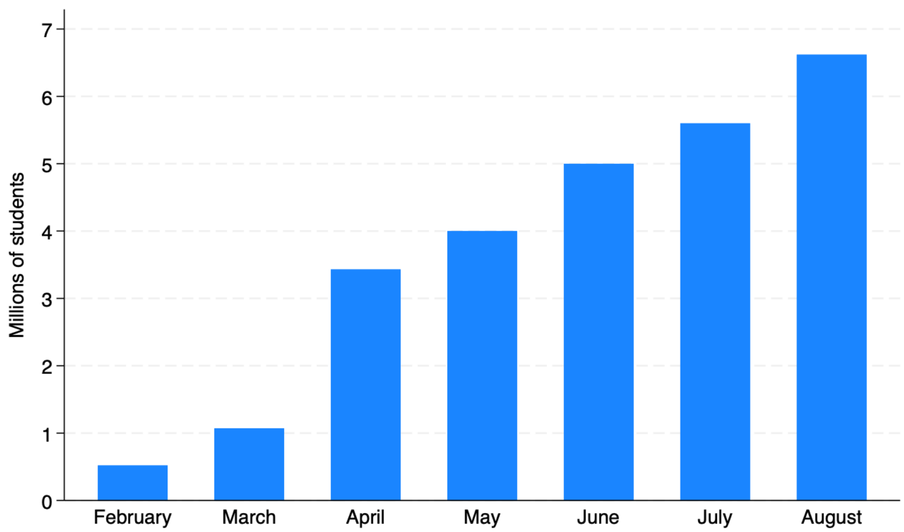
Ulyssea, G. (2018). Firms, Informality, and Development: Theory and Evidence from Brazil. *American Economic Review*, 108(8):2015–47.

Ulyssea, G. (2020). Informality: Causes and Consequences for Development. *Annual Review of Economics*, 12(1):525–546.

Ulyssea, G., Bobba, M., Gadenne, L., and Harari, M. (2025). Informality. *VoxDevLit*, 6(2).

A Additional Material

Figure A.1: Evolution of cumulative students' enrollment to social insurance, 2016

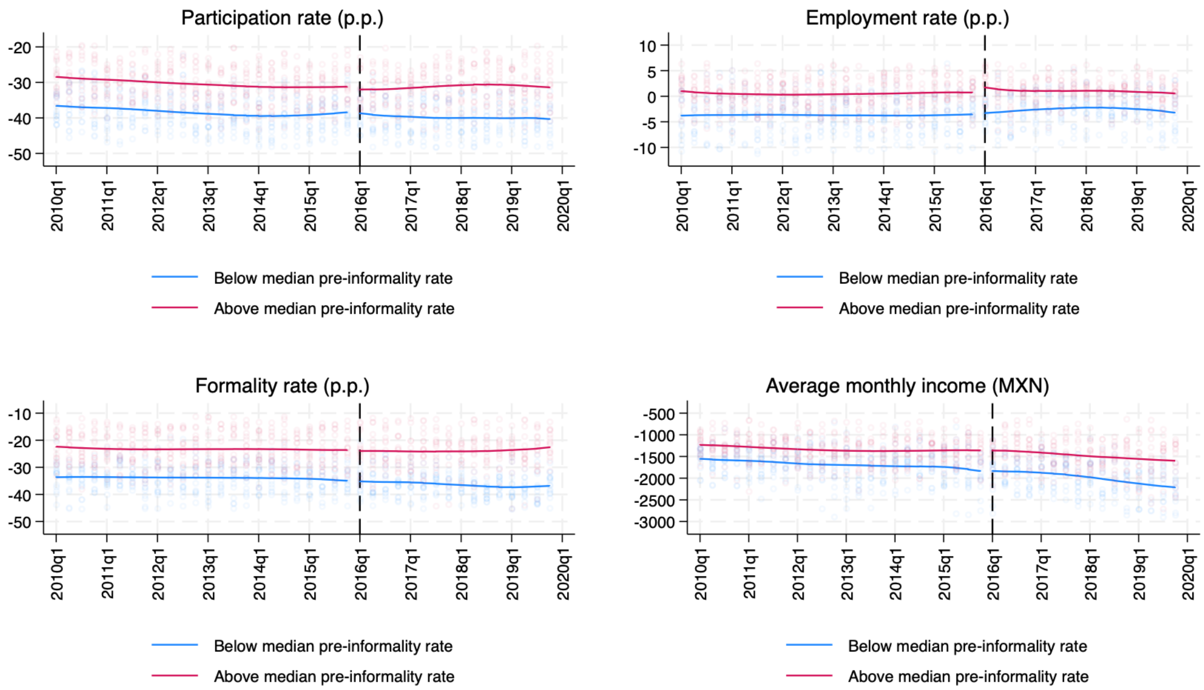


Source: IMSS. *Informe de Labores y Programa de Actividades 2016–2017.*

Figure A.2: Regions of Mexico

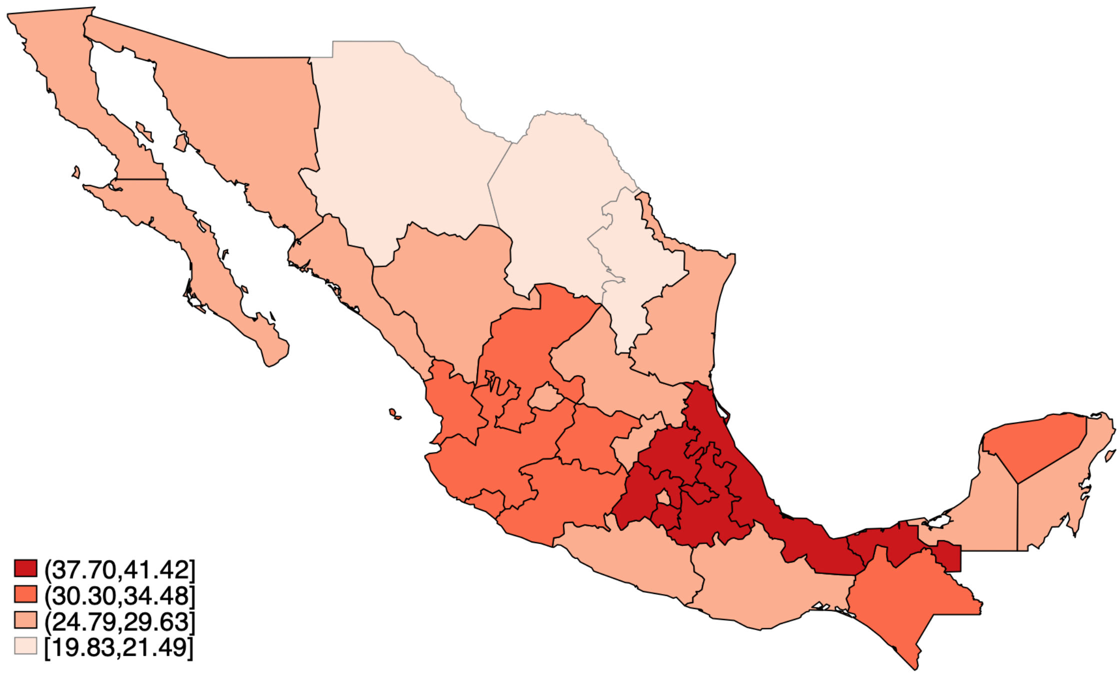
Source: Regiones de México. Northwest region includes Baja California, Baja California Sur, Chihuaha, Durango, Sinaloa and Sonora. Northeast region includes Coahuila, Nuevo León and Tamaulipas. West region includes Colima, Jalisco, Michoacán and Nayarit. East region includes Hidalgo, Puebla, Tlaxcala and Veracruz. Central north region includes Aguascalientes, Guanajuato, Querétaro, San Luís Potosí and Zacatecas. Central south region includes Mexico City, Mexico State and Morelos. South west region includes Chiapas, Guerrero, and Oaxaca. Southeast region includes Campeche, Quintana Roo, Tabasco and Yucatán.

Figure A.3: Long differences trends (difference between 15–18 and 21–24 cohorts by state)



Note: Quarterly long differences are computed for each state as the outcome gap between ages 15–18 (treated) and 21–24 (control). States are split at the median pre-policy informality (2013–2015), and lines plot LOWESS-smoothed trends pre/post-2016. Table A.6 reports the corresponding formal check: baseline long-difference estimates conditioning on each state's cross-cohort gaps three years before the policy.
Source: Author's calculations with data from ENOE.

Figure A.4: Alternative definition of pre-policy informality rate by state (%), only employees
2013–2015 average

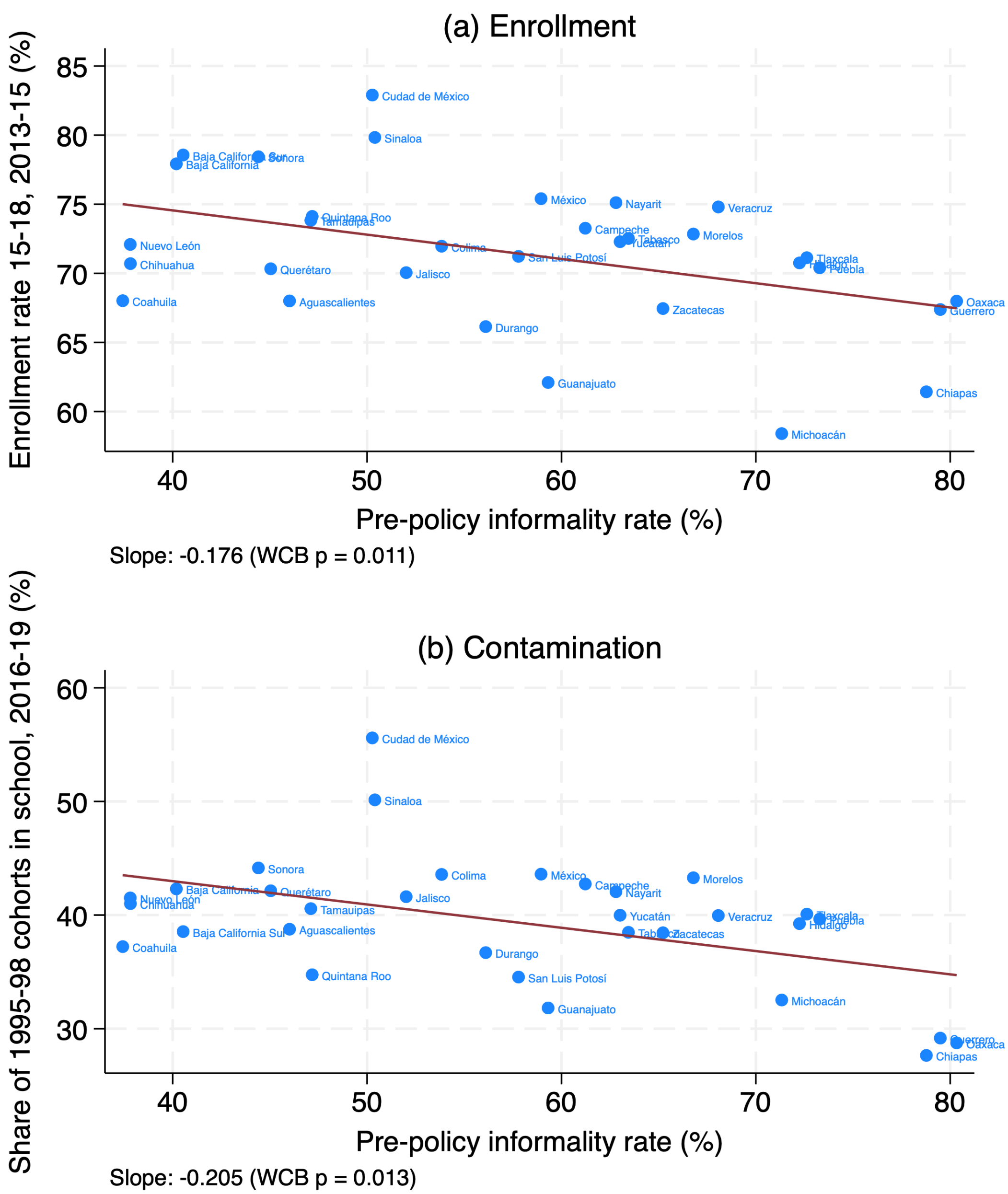


Source: Author's calculation using data from ENOE.

Figure A.5: Public share of upper-secondary enrollment, 2015–2016 (%)

Source: SEP. *Principales Cifras del Sistema Educativo Nacional 2015-2016.*

Figure A.6: Orthogonality of exposure ingredients to pre-policy informality



Note: Each point is a state. Panel (a) plots the pre-policy school-attendance rate of 15–18 year-olds (ENOE, 2013–2015) against the pre-policy informality rate; panel (b) plots the share of the 1995–1998 cohorts observed attending school during 2016–2019. Lines are unweighted linear fits; slopes and wild cluster bootstrap p-values are reported in Table A.7, Panel B.
Source: Author's calculations using data from ENOE.

Table A.1: Descriptive Statistics

	Long Differences (2019)		Panel Estimates (2016-2019)	
	(1) Exposed cohorts	(2) Unexposed cohorts	(3) Exposed cohorts	(4) Unexposed cohorts
Participation rate	0.23	0.61	0.21	0.53
Employment rate	0.93	0.93	0.95	0.92
Formality rate	0.08	0.41	0.07	0.35
Monthly income	3,501	5,384	2,906	4,602
School enrollment	0.77	0.31	0.79	0.40
Pre-policy informality rate	0.59	0.58	0.59	0.58
Log GDP per capita in 2013	11.68	11.72	11.67	11.70
Educational attainment in 2015	9.10	9.18	9.10	9.16
Age	16.05	22.02	15.60	20.48
Men	0.50	0.50	0.49	0.50
Formal household head	0.48	0.50	0.47	0.49
Northwest region	0.13	0.13	0.13	0.14
Northeast region	0.10	0.10	0.09	0.10
West region	0.12	0.12	0.12	0.12
East region	0.15	0.15	0.15	0.15
Central north region	0.12	0.12	0.12	0.12
Central south region	0.21	0.22	0.21	0.22
Southwest region	0.11	0.09	0.11	0.10
Southeast region	0.06	0.06	0.06	0.06
Observations	128	128	5,155	3,456

Note: Mean values for each variable are reported, weighted by survey weights. Columns (1) and (2) correspond to the long differences estimates reported in Table 2. Columns (3) and (4) correspond to the panel estimates reported in Table 3. For the long differences estimates, exposed cohorts only include fully exposed cohorts; for the panel estimates exposed cohorts include partially exposed cohorts as defined by the exposure measure plotted in figure 1. Rates (participation, employment, formality, school enrollment, pre-policy informality) are displayed as proportions (0–1) in this table. In all regression specifications the pre-policy informality rate and labor market outcome rates are scaled as percentage points (0–100); regression coefficients are therefore interpreted as the effect of a 1-percentage-point change in the respective variable.

Table A.2: Population-Based Decomposition of Cross-Cohort Long Differences, 2019

	(1) Employment rate	(2) Formal employment	(3) Informal employment	(4) Unemployment	(5) Non-participation
Panel A					
Pre-policy informality rate	0.49***	0.73***	−0.24***	0.02	−0.51***
	(0.05)	(0.04)	(0.05)	(0.02)	(0.05)
	[<0.001]	[<0.001]	[<0.001]	[0.284]	[<0.001]
Panel B					
Pre-policy informality rate	0.61***	0.90***	−0.29**	−0.02	−0.59***
	(0.08)	(0.09)	(0.08)	(0.04)	(0.08)
	[<0.001]	[<0.001]	[0.021]	[0.673]	[<0.001]
Log GDP per capita in 2013	7.12***	4.47***	2.65**	−1.16	−5.96***
	(0.89)	(1.27)	(1.05)	(0.86)	(1.02)
Educational attainment in 2015	−3.01**	0.94	−3.95***	−0.24	3.25***
	(1.20)	(1.39)	(1.25)	(0.54)	(1.16)
Regional controls	✓	✓	✓	✓	✓
Quarter FE	✓	✓	✓	✓	✓
Mean dependent variable	−34.55	−19.98	−14.57	−3.26	37.81
Observations	128	128	128	128	128
Panel C: formal and informal employment by type of job					
		Formal, wage	Formal, self-emp. & other	Informal, wage	Informal, self-emp. & other
Pre-policy informality rate		0.89***	0.01	−0.32***	0.03
		(0.09)	(0.01)	(0.07)	(0.06)
		[<0.001]	[0.244]	[0.008]	[0.720]

Note: All dependent variables are exposed-minus-unexposed cohort gaps in shares of the cohort population (in percentage points), so that columns (2)–(5) sum to column totals by construction: formal employment, informal employment, unemployment, and non-participation partition the population, and the coefficients in columns (2)–(5) of each panel sum to zero. Column (1) is the employment-to-population ratio, the sum of columns (2) and (3). Panel C splits formal and informal employment into subordinate remunerated (wage) employment and self-employment and other positions, estimated with the Panel B specification; the wage and self-employment coefficients sum to the corresponding aggregate in Panel B. Sample, cell structure, weights (square root of cell population), and estimation (weighted least squares, standard errors clustered at the state level in parentheses, wild cluster bootstrap p-values for the pre-policy informality coefficient in brackets) are identical to Table 2. Baseline 2019 levels (percent of cohort population): employment-to-population 19.4 (exposed) and 54.0 (unexposed); formal employment 1.3 and 21.3; informal employment 18.2 and 32.7; unemployment 1.4 and 4.7; non-participation 79.2 and 41.4. ***p<0.01, **p<0.05, *p<0.1

Table A.3: Robustness of Inference and Leverage

	Long differences		Panel	
	Participation	Formality	Participation	Formality
Coefficient	0.59	0.87	0.44	1.28
Panel A: standard errors / p-values				
Cell bootstrap	(0.07)	(0.11)	—	—
Cluster-robust, state (CRV1)	(0.09)	(0.16)	(0.05)	(0.06)
Block bootstrap, states	(0.14)	(0.21)	—	—
Cluster jackknife (CV3)	(0.12)	(0.21)	(0.07)	(0.06)
Wild cluster bootstrap p-value	<0.001	0.003	<0.001	<0.001
Wild bootstrap 95% CI	[0.40, 0.81]	[0.44, 1.24]	[0.30, 0.56]	[1.16, 1.45]
State-collapsed (32 obs.)	0.59 (0.10)	0.87 (0.18)	—	—
Panel B: leverage				
Leave-one-state-out range	[0.56, 0.67]	[0.79, 0.98]	[0.40, 0.48]	[1.24, 1.31]
Excluding Chiapas, Guerrero and Oaxaca	0.61	0.88	0.46	1.29
Panel C: weighting of cells				
$\sqrt{\text{population}}$ (baseline)	0.59	0.87	0.44	1.28
Population	0.59	0.91	0.45	1.29
Unweighted	0.60	0.85	0.43	1.27

Note: All rows refer to the coefficient on pre-policy informality (long differences, Table 2 Panel B specification) or on pre-policy informality interacted with exposure (panel, Table 3 specification). Point estimates are identical across inference methods by construction. Wild cluster bootstrap uses Webb weights with $9,999$ replications, clustering by state (Roodman et al., 2019); CV3 jackknife standard errors follow MacKinnon et al. (2023). The state-collapsed specification averages the four 2019 quarters within each state (population-weighted) and drops quarter fixed effects. Panel C reports coefficients under alternative regression weights; all remain significant at the 1 percent level by wild cluster bootstrap.

Table A.4: Panel Estimates Robustness to State Linear Trends, 2016–2019

	(1) Participation rate	(2) Employment rate	(3) Formality rate	(4) Monthly income
Pre-policy informality	0.45***	0.15***	1.29***	24.52***
rate × Exposure	(0.06)	(0.03)	(0.06)	(8.29)
Log GDP per	2.05	0.44	3.73**	-64.72
capita in 2013 × Exposure	(1.05)	(1.08)	(1.05)	(131.27)
Educational attainment	-0.32	1.85**	1.92	-58.32
in 2015 × Exposure	(0.71)	(0.58)	(1.29)	(101.38)
Cohort FE	✓	✓	✓	✓
State FE	✓	✓	✓	✓
Quarter FE	✓	✓	✓	✓
State linear time trend	✓	✓	✓	✓
Mean dependent variable	33.31	93.84	17.99	4,299.21
Observations	5,094	5,091	5,091	5,046

Note: Standard errors clustered by state in parentheses. The square root of cell sizes are used as weights for the observations. Cell sizes are identical across specifications, except for the income one as I restrict to individuals reporting strictly positive income.
***p<0.01, **p<0.05, *p<0.1

Table A.5: Robustness to Informality-Specific Cohort Trends and Age Profiles

	(1) Baseline 1995–2004	(2) Baseline 1991–2004	(3) + linear trend	(4) + quadratic trend	(5) + age profile
Panel A: participation rate					
Informality × Exposure	0.44***	0.39***	0.15*	0.22***	0.15**
	(0.05)	(0.04)	(0.07)	(0.07)	(0.06)
Informality × cohort trend			0.03***	0.04***	
			(0.01)	(0.01)	
Informality × cohort trend^2				−0.002**	
				(0.001)	
Informality × age					−0.03***
					(0.01)
WCB p-value (exposure)	<0.001	<0.001	0.052	0.006	0.039
Observations	5,094	7,142	7,142	7,142	7,142
Panel B: formality rate					
Informality × Exposure	1.28***	1.21***	1.09***	0.78***	0.93***
	(0.06)	(0.06)	(0.08)	(0.07)	(0.07)
Informality × cohort trend			0.02	−0.04**	
			(0.01)	(0.02)	
Informality × cohort trend^2				0.007***	
				(0.001)	
Informality × age					−0.04**
					(0.01)
WCB p-value (exposure)	<0.001	<0.001	<0.001	<0.001	<0.001
Observations	5,091	7,139	7,139	7,139	7,139

Note: All columns estimate the Table 3 panel specification (cohort, state, and quarter fixed effects; GDP and education controls interacted with exposure; square-root-of-cell-population weights; 2016–2019). Column (1) uses the Table 3 cohort range (1995–2004); columns (2)–(5) extend the sample to cohorts 1991–2004, adding four untreated cohorts that help separate smooth cohort profiles from the exposure step. Columns (3)–(5) add interactions of pre-policy informality with a linear cohort trend, a quadratic cohort trend, and age (year minus cohort), respectively, so the exposure coefficient is identified from the discrete exposure ramp net of smooth informality-specific profiles. Because exposure ramps from zero to one over four cohorts, flexible trends absorb part of the ramp by construction; these estimates are therefore conservative bounds. Standard errors clustered by state in parentheses; wild cluster bootstrap p-values for the exposure coefficient reported (Webb weights, 9,999 replications). ***p<0.01, **p<0.05, *p<0.1

Table A.6: Long Differences Conditioning on Pre-Policy Cohort Gaps, 2019

	(1) Participation rate	(2) Employment rate	(3) Formality rate	(4) Monthly income
Panel A				
Pre-policy informality rate	0.40***	0.10***	0.52***	28.09***
	(0.06)	(0.02)	(0.07)	(4.14)
	[<0.001]	[0.001]	[<0.001]	[<0.001]
Cohort gap in 2013	0.24	0.31***	0.46***	0.10
(same outcome)	(0.15)	(0.08)	(0.11)	(0.08)
Observations	128	128	128	128
Panel B				
Pre-policy informality rate	0.63***	0.12*	0.58**	11.02
	(0.10)	(0.05)	(0.18)	(9.98)
	[<0.001]	[0.072]	[0.017]	[0.280]
Cohort gap in 2013	-0.08	0.24**	0.42***	-0.01
(same outcome)	(0.12)	(0.09)	(0.11)	(0.08)
Log GDP per	5.98***	1.47*	0.63	-137.07
capita in 2013	(0.88)	(0.86)	(1.25)	(128.31)
Educational attainment	-3.16***	-0.24	1.70	-333.14**
in 2015	(1.07)	(0.69)	(1.55)	(142.36)
Regional controls	✓	✓	✓	✓
Quarter FE	✓	✓	✓	✓
Observations	128	128	128	128

Note: Each column reports the Table 2 long-difference specification augmented with the state's cross-cohort gap in the same outcome measured in 2013, three years before the start of the policy. The 2013 gap is the difference between the 15-18 and 21-24 age groups by state and quarter (the construction underlying figure A.3), matched to each 2019 cell by state and quarter of the year. Standard errors clustered by state in parentheses; wild cluster bootstrap p-values for the pre-policy informality coefficient in brackets (Webb weights, 9,999 replications). Cells are weighted by the square root of the cell population (the square root of the positive-income cell population in column (4), which restricts to individuals reporting strictly positive income).
***p<0.01, **p<0.05, *p<0.1

Table A.7: Predicted Exposure and Exposure Ingredients

Panel A: panel estimates under predicted exposure						
	Formality rate			Participation rate		
	(1) Baseline	(2) Enrollment	(3) Enroll.× public	(4) Baseline	(5) Enrollment	(6) Enroll.× public
Informality × exposure measure	1.28***	1.50***	1.59***	0.44***	0.51***	0.56***
	(0.06)	(0.07)	(0.12)	(0.05)	(0.05)	(0.08)
	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]
Observations	5,091	5,091	5,091	5,094	5,094	5,094

Panel B: orthogonality of exposure ingredients to pre-policy informality		
	Enrollment rate 15–18, 2013–15	Contamination 1995–98 cohorts
Pre-policy informality rate	−0.18**	−0.21**
	(0.06)	(0.07)
	[0.011]	[0.013]
R-squared	0.19	0.22
Observations	32	32

Note: Panel A re-estimates the Table 3 panel specification replacing cohort exposure with predicted exposure: cohort exposure scaled by the state's pre-policy school-attendance rate of 15–18 year-olds (columns (2) and (5); ENOE 2013–2015, mean 71.5 percent) and additionally by the public share of upper-secondary enrollment (columns (3) and (6); SEP, *Principales Cifras del Sistema Educativo Nacional 2015–2016*, mean 81 percent). Controls are interacted with the same exposure measure as the treatment variable in each column. Panel B regresses each exposure ingredient on the pre-policy informality rate across the 32 states; contamination is the share of the 1995–1998 cohorts observed attending school during 2016–2019. Standard errors clustered by state in parentheses (robust in Panel B); wild cluster bootstrap p-values in brackets (Webb weights, 9,999 replications). The implied exposure differential between designated exposed and unexposed cohorts lies between 0.58 (enrollment × public share) and 0.18 (fully counting college contamination of the comparison group), so effects per enrolled-eligible student are at least 1.7 times the intention-to-treat estimates.
***p<0.01, **p<0.05, *p<0.1

B Operational Details of the School-Linked Social Insurance Expansion

- **Enrollment channel (2016 cohort): assisted mass enrollment.** In early 2016, federal and state authorities coordinated with public high schools to batch-enroll students. Schools transmitted rosters identified by CURP (the national population registry code), which IMSS used to assign or validate the student's permanent social security number (NSS). This step was largely automatic from the student's perspective. No individual application was required to be assigned/validated an NSS in 2016 for those on school rosters.

- **Access to services (activation/clinic choice): minor residual steps.** To actually use healthcare services, students typically needed to confirm or choose a primary clinic (UMF) and, in some cases, complete a brief registration/verification. These are low-stakes, one-off steps (often handled by schools or the student) and do not reverse eligibility.

- **Cost reduction relative to status quo ante.** Before 2016, obtaining an NSS as a youth without a formal job involved an application and document gathering; the 2016 campaign eliminated that transaction cost at scale by pre-assigning/validating the NSS off administrative records (via CURP and school rosters).

- **Post-2019 practice (for context only)**. Later cohorts generally request their NSS individually (no longer batch-assigned), which is also why my empirical window centers on 2016–2019.

Figure B.1: "Tienes IMSS" campaign flyer

Source: IMSS. Official Twitter account (April 18, 2016).

Figure B.2: Political event announcing the affiliation of students to IMSS in Mexico City

Source: IMSS. *El Seguro Médico para estudiantes es un escudo para prevenir enfermedades crónicas: Mikel Arriola.*